



A fast and robust third medium contact approach using the neighbored element method

Max von Zabiensky^{ID*}, Dustin R. Jantos^{ID}, Philipp Junker^{ID}

Leibniz University Hannover, Institute of Continuum Mechanics, Hannover, Germany

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ABSTRACT

In several engineering applications, self-contact is a major effect or even desired feature, e.g., energy absorption in severe structural deformations, gripper in (soft-)robotics, and the optimization of those. Usual contact approaches require extensive calculation effort, but a fast and robust simulation approach could lead to remarkable improvements in product design. The third medium contact is a very promising approach, which can fulfill the requirements mentioned above. This method was also already applied in topology optimization approaches. However, the most concurrent regularization approaches require at least quadratic shape functions for hexahedron finite elements and at least cubic shape functions for tetrahedrons. In recent works, a new regularization technique was developed: herein, the gradients of deformation measures are computed by additional nodal degrees of freedom in the finite elements, i.e., as additional fields of unknowns. This is accompanied by a reduction of the order of the shape function. In this work, we derive a special discretization of the additional unknowns which enables self-contact based on the neighbored element method. This approach leads to a fast algorithm for the computation of the regularized set of partial differential equations: the displacements are the only nodal degrees of freedom and can be computed with linear finite element shape functions while the additional unknowns are discretized in the quadrature points of the mesh and solved in a staggered manner. The results are critically analyzed and illustrated for several two-dimensional problems.

1. Introduction

The modeling of contact mechanics goes a long way in the field of finite element methods, since the very first formulation in [1]. Until today, the applications of advanced contact formulations attain significant importance in the real world, see [2]. Based on the applied contact method, the complexity of the numerical treatment may vary strongly. The simulation of a contact problem goes along with local and global search algorithms, i.e., [3,4], as well as a specific handling of variational inequalities, due to the nature of the formulation of these problems, i.e., [5,6]. One possible way to reduce the computation effort for the detection of potential contact zones is to preassign the contact regions, like node-to-node contact, node-to-segment contact, or the mortar method. For more details and an overview on these methods, interested readers are referred to [7,8]. To handle the arising inequalities, many formulations where investigated over decades, like penalty method, barrier functions, or augmented Lagrangian techniques. For further details, see [7,9,10]. In [11], these approaches have also been extended to finite deformations. Even a meshfree approach, based on a combination of the third medium contact with the isogeometric meshfree coupling, was developed in [12]. One famous contact formulation for large deformations is developed in [13], whereby the contact area is discretized by nodes and segments.

* Corresponding author.

E-mail address: von-zabiensky@ikm.uni-hannover.de (M. von Zabiensky).

The connection between these nodes and segments is based on a master–slave condition and is applied by a single pass algorithm. This approach was further developed by [14,15], which introduced a two-pass algorithm to fulfill the 2D contact patch test. Several developments followed this idea and applied further discretization approaches. Many of them refer to the mortar method, which originated from the field of domain decomposition methods, see [16–18]. The idea of the domain decomposition was extended in [19], by defining patches between the contact segments, which results from a triangulation of the intermediate domain. Within the patches the authors embedded a displacement field of the same order as the contacting bodies. Therewith, the Lagrange multipliers and strain-like measurements are defined along the whole domain, rather on the contact segments only. Founding on this approach in [19], the authors enabled the computation of various contact problems, shown in their following work [20,21].

Out of these approaches there is one rather simple scheme, which was first introduced in [22], where anisotropic modeling was used in the contact area to avoid inequality constraints. This so called third medium contact method (TMC) is based on the idea, that the contact is implemented by a filling mesh between the solid bodies, i.e., discretizing void regions between potential contact faces. It is worth mention that the TMC method and the contact domain method in [19] seem to share significant similarities since both introduce a finite discretization for the intermediate domain by numerical elements, respectively patches. However, it must be emphasized that there is a major difference due to discretization: in case of the contact domain method, the contact area is pre-described on each body and then connected with a single layer of triangulated patches. In contrast, in the TMC method, a volumetric mesh is applied between potential contact areas which allow for more regular meshes even for very large initial gaps potentially improving numerical accuracy and stability due to improved mesh quality. In addition, meshing of the overall problem can be achieved by usual meshing algorithms by assuming the “void” simply to be a second material phase. Thus, it is not necessary to explicitly pre-describe discretized contact points and/or faces, but instead to stabilize highly deformed elements. This can be achieved with an additional regularization without introducing a residual stiffness for void elements except for the case of contact. This approach was further refined over the last years, see [23–27], whereby the regularization is applied to the gradient of the deformation gradient. With these higher gradients, higher deformation modes are included in the theory, and in turn also higher-order shape functions within the finite element formulation. An alternative stabilization was achieved in [25], whereby the Laplacian of the deformation field was taken into account. As a result, a snap-through problem could be simulated. Further regularization approaches were developed: in [28], the polar decomposition of the deformation gradient is used to regularize the modeling of pneumatic structures. Moreover, the TMC was used to optimize non-linear thermo-mechanical problems in [29]. Last but not least, a friction model was presented in [26], where the TMC follows the crystal plasticity theory.

These mentioned applications established the TMC in stabilizing finite element simulations with contact and yet, there is more potential. Especially in the field of optimization problems [30], or (soft) robotics [31], is a remarkable need for a stabilization of void phases and enabling the simulation of effects like (self-)contact, snapping, or buckling, while the computational costs stay moderate. A most recent improvement of the method is given in [32], wherein the regularization of the third medium is based on the gradient of the rotation part of the deformation gradient as well as the Jacobian, i.e., volumetric deformation. Therefore, a reduced, but very robust implementation was developed, which only takes the relevant modes of the gradient of the deformation gradient into account. Further, it was proven that a reduced formulation with finite elements with linear shape functions which are a basic concept of the finite discretizations [33], can be applied. However, additional field variables have to be introduced, which are treated as additional nodal degrees of freedom. Furthermore, this approach was applied to 3D problems [34] and a stabilization-free virtual element method for 2D problems [35].

In this work, we pick up on the reformulation of the regularization from [32] to provide a contact simulation in which the additional field values of the regularization are solved in a special numerical scheme: we apply the neighbored element method (NEM) which is a staggered approach solving the additional degrees of freedom via a (meshfree) finite difference scheme while the displacements are the only nodal degrees of freedom in the finite element approach. In this work, we discretize the partial differential equations regarding the regularization within the quadrature points of the finite element mesh used to compute the displacement field. As a consequence, a remarkable reduction of the computation time can be achieved, likewise to the work of [36], while achieving maximum spatial resolution for the additional field variables. All examples in this paper will be solved with linear, i.e., first order, shape functions which results in a fast and efficient computation of the TMC. With numerical examples and benchmark problems, it will be illustrated how the new solving approach effects the interaction of the solid and void phases.

The paper is structured as follows: in Section 2 the variational formulation for the TMC is explained. In Section 3 the novel numerical approach via NEM is explained and numerical results are given in Section 4.

2. Methodology

2.1. Preliminaries

For the formulation of the TMC, it is necessary to define a medium as an infill between potential contact zones of the solid regions, which we refer to as void phase. This void medium has to be highly compliant, but as soon as the solid phases are about to get into contact, it has to get stiffer to act like a barrier and to enable contact forces. Of course, the stiffness of this void phase has to be effective enough to prevent penetration and provide a small as possible gap in case of contact, but should also be as compliant as possible prior to contact to not impair contact free deformations. Beside this compromise, it should be noted that strongly compressed or distorted void elements should not affect the stability of the numerical treatment.

2.2. Continuum model

The kinematic relations have to be formulated first based on the initial configuration. Regarding to the mapping between the initial and current configuration, we define the deformation gradient $\mathbf{F}$ based on the gradient of the deformation field $\nabla \mathbf{u}$. From this, the Cauchy–Green deformation tensor $\mathbf{C}$, as well as the Green–Lagrange deformation tensor $\mathbf{E}$ can be derived:

$$\mathbf{F} = \mathbf{I} + \nabla \mathbf{u} \quad \mathbf{E} = \frac{1}{2} (\mathbf{F}^T \mathbf{F} - \mathbf{I}) \quad \mathbf{C} = 2\mathbf{E} + \mathbf{I}. \quad (1)$$

Detailed derivations can be found in [37,38]. To simulate finite deformations of the solid phase, the strain energy is used analogously to [32], which is adapted here to the two-dimensional case. This formulation describes a hyperelastic material behavior with a split in volumetric and deviatoric terms:

$$\Psi_M(\mathbf{C}) = \frac{K}{2} [\ln(J)]^2 + \frac{\mu}{2} (J^{-1} \text{tr}(\mathbf{C}) - 2). \quad (2)$$

The engineering parameters which characterize the material behavior are described by the shear modulus μ and the compression modulus K . The Jacobian is given by $J = \det(\sqrt{\mathbf{C}})$ and the trace of the Cauchy–Green deformation tensor is defined by $\text{tr}(\mathbf{C})$. In general, it is also possible to apply other (inelastic) material models for the solid phase, since the stabilization is integrated only into the void phase which will be shown later. However, it is clear that the stabilization requires the computation based on the finite deformation theory. For the formulation of the contact problem, the third medium approach according to [32] is used, introducing the distortion energy for the two-dimensional case as

$$\Psi_V(\mathbf{C}) = \gamma \frac{\mu}{2} (J^{-1} \text{tr}(\mathbf{C}) - 2). \quad (3)$$

The parameter γ scales the magnitude of the energy in comparison to the hyperelastic non-void material phase, i.e., the effective stiffness of void material. It must be chosen small enough so that the solid material does not suffer any influence from the third medium in non-contact regions. Since the phases of the third medium and the solid material are to be clearly separated in the undeformed configuration, the parameter χ is introduced, which indicates the solid material for $\chi = 1$ and the void material phase for $\chi = 0$. The total elastic strain energy for the problem body Ω containing the actual structure and the void regions is thus defined by

$$\Psi = \chi \Psi_M + (1 - \chi) \Psi_V \quad \forall \mathbf{X} \in \Omega. \quad (4)$$

2.3. Formulation of the regularization

By defining the strain energy of the third medium, the elastic material behavior of the void phase is specified. However, this formulation alone is not yet sufficient, since the elements tend to strong distortions at large deformation states, which does not entail a stable calculation of finite deformations [23]. For this reason, a regularization for the void material ($\chi = 0$) is necessary, which in this work is based on the regularization of the gradient of the rotational and volumetric part of the deformation gradient, originally introduced in [32]. Here, only the gradient of the rotational part $\nabla \mathbf{R}$ is accounted for. In [32], it has already been shown that the regularization due to the deformation gradient can be neglected by using linear shape functions.

The calculation of the rotation can be derived from the polar decomposition of the deformation gradient $\mathbf{F} = \mathbf{R} \cdot \mathbf{U}$. By converting this formulation according to the right stretch tensor $\mathbf{U} = \mathbf{R}^T \cdot \mathbf{F}$, the rotation angle φ for the two-dimensional case can be calculated using the components of the deformation gradient. The symmetry of the right stretch tensor $U_{12} = U_{21}$ can be utilized yielding

$$\tan(\varphi(\mathbf{F})) = \left[\frac{F_{12} - F_{21}}{F_{11} + F_{22}} \right]. \quad (5)$$

Furthermore, it can be shown that the squares of the norms of the gradient of the rotation tensor and the angle of rotation are equal $\|\nabla \mathbf{R}\|^2 = \|\nabla \varphi\|^2$, cf. [32].

However, the rotational gradient, or more precisely the angle φ , will not be regularized directly in the proposed method. Instead, an additional field variable p is introduced, which is subject to regularization and coupled to the rotation angle φ by means of a square error minimization. Such kind of regularization approach was already applied in gradient enhanced damage modeling in [39] and topology optimization [40]. For further details regarding the specific formulation of TMC, we refer to [32].

For the regularization, the following additional term for the strain energy is introduced:

$$\Psi_R(\mathbf{F}) = \frac{\gamma}{2} [\beta_p (\tan \varphi(\mathbf{F}) - p)^2 + \alpha_p \|\nabla p\|^2]. \quad (6)$$

The regularization will only be applied in the void phase. The penalty parameters β_p and α_p are intended to control the coupling of the variable p to the rotation gradient and the potency of the regularization, respectively. To the end, this will achieve a spatial smoothing of $\tan(\varphi(\mathbf{F}))$.

Likewise, the regularization for the Jacobian $J = \det(\mathbf{F})$ will be implemented as a additional term for the strain energy:

$$\Psi_J(\mathbf{F}) = \frac{\gamma}{2} [\beta_q (J - q)^2 + \alpha_q \|\nabla q\|^2]. \quad (7)$$

Herein, the penalty parameters β_q and α_q are controlling the coupling of the field variable q to the gradient of the Jacobian. Therewith, a spatial smoothing of the volumetric inflation will be achieved. Due to the spatial smoothing, severe distortions are

prevented in the void elements which can lead to numerical instabilities. Furthermore, the formulation of the strain energy in Eqs. (6) and (7) only adding energy to the system, even if the Jacobian J , or the rotation $\tan(\varphi(\mathbf{F}))$ take negative values. In addition, the reciprocal Jacobian in Eq. (3) ensures an incompressible martial behavior whereby a self-penetration of the void elements is prevented.

In contrast to the solution in [32], the numerical solution of the resulting differential equations for the solution of p and q will be completely different in this work and is based on its strong form. The differences of the numerical solution schemes are presented in detail in Section 3.

2.4. Stationary conditions

Using the strain energies, see (2), (3), (6) and (7), the Gibbs energy is formulated as

$$\mathcal{G} = \int_{\Omega} [\chi \Psi_M + (1 - \chi) \Psi_V + (1 - \chi) \Psi_R + (1 - \chi) \Psi_I] dV - \int_{\Omega} \mathbf{b} \cdot \mathbf{u} dV - \int_{\partial\Omega} \mathbf{t} \cdot \mathbf{u} dA. \quad (8)$$

Here, $\mathbf{b}$ and $\mathbf{t}$ denote the body forces and the traction forces on the Neumann boundary of the solid material, respectively. Since the strain energies are split into their respective phases by the parameter χ , they can be related to the entire domain Ω . The governing equations are obtained from the principle of virtual work, i.e.,

$$\mathcal{G} \rightarrow \text{stat} \quad \Rightarrow \quad \delta \mathcal{G} = \delta_{\mathbf{u}} \mathcal{G} + \delta_p \mathcal{G} + \delta_q \mathcal{G} = 0 \quad \forall \delta \mathbf{u}, \delta p, \delta q. \quad (9)$$

The stationary condition of the variation of the Gibbs energy according to the displacement $\mathbf{u}$ reads

$$\delta_{\mathbf{u}} \mathcal{G} = \chi \int_{\Omega} \frac{\partial \Psi_M}{\partial \mathbf{E}} : \delta \mathbf{E} dV + (1 - \chi) \left(\int_{\Omega} \frac{\partial \Psi_V}{\partial \mathbf{E}} : \delta \mathbf{E} dV + \int_{\Omega} \frac{\partial \Psi_R}{\partial \mathbf{F}} : \delta \mathbf{F} dV + \int_{\Omega} \frac{\partial \Psi_I}{\partial \mathbf{F}} : \delta \mathbf{F} dV \right) = 0 \quad \forall \delta \mathbf{u}. \quad (10)$$

The variation of the Gibbs energy according to the field variable p includes only the regularization part (6) of the strain energy and yields

$$\delta_p \mathcal{G} = (1 - \chi) \gamma \left[\int_{\Omega} -\beta_p (\tan \varphi(\mathbf{F}) - p) \delta p dV + \int_{\Omega} \alpha_p \nabla p \cdot \nabla \delta p dV \right] = 0 \quad \forall \delta p. \quad (11)$$

For purpose of numerical discretization via finite difference scheme within the NEM (c.f. Section 3), we reformulate Eq. (11) to its strong form by applying an integration by parts and Gauss's integral theorem:

$$\delta_p \mathcal{G} = (1 - \chi) \gamma \left[\int_{\Omega} -\beta_p (\tan \varphi(\mathbf{F}) - p) \delta p dV + \int_{\partial\Omega} \alpha_p \mathbf{n} \cdot \nabla p \delta p dV - \int_{\Omega} \alpha_p \nabla^2 p \delta p dV \right] = 0 \quad \forall \delta p \quad (12)$$

with the Laplace operator $\nabla^2 p := \nabla \cdot \nabla p$. The gradient ∇p in Eq. (12) is now only applied on the boundary $\partial\Omega$ and yields homogeneous Neumann boundary condition. Since the stationary condition has to be fulfilled independently on the boundary $\partial\Omega$ and the inner domain Ω , we find the strong form as

$$\begin{aligned} -\beta_p (\tan \varphi(\mathbf{F}) - p) - \alpha_p \nabla^2 p &= 0 & \forall X \in \Omega \\ \mathbf{n} \cdot \nabla p &= 0 & \forall X \in \partial\Omega \end{aligned} \quad (13)$$

The last variation of the Gibbs energy according to the field variable q reads

$$\delta_q \mathcal{G} = (1 - \chi) \gamma \left[\int_{\Omega} -\beta_q (J - q) \delta q dV + \int_{\Omega} \alpha_q \nabla q \cdot \nabla \delta q dV \right] = 0 \quad \forall \delta q. \quad (14)$$

Analogously to the field variable p , we reformulate Eq. (14) to its strong form, in the same manner as shown for the other field variable, i.e.,

$$\delta_q \mathcal{G} = (1 - \chi) \gamma \left[\int_{\Omega} -\beta_q (J - q) \delta q dV + \int_{\partial\Omega} \alpha_q \mathbf{n} \cdot \nabla q \delta q dV - \int_{\Omega} \alpha_q \nabla^2 q \delta q dV \right] = 0 \quad \forall \delta q \quad (15)$$

and thus,

$$\begin{aligned} -\beta_q (J - q) - \alpha_q \nabla^2 q &= 0 & \forall X \in \Omega \\ \mathbf{n} \cdot \nabla q &= 0 & \forall X \in \partial\Omega. \end{aligned} \quad (16)$$

The numerical treatment is presented in Section 3.

3. Numerical treatment

3.1. Preliminaries

Eq. (10) can be evaluated with a classical non-linear Finite Element Method (FEM) applying a Newton–Raphson iteration over multiple load steps [41]. The displacements are approximated in this work with linear shape functions, i.e., first-order spatial discretization.

To calculate the field variables p and q , we use the strong form in Eqs. (13) and (16). We apply a generalized Finite Difference Method (FDM), which requires the neighbored relations of a discretized cloud of points which we choose to be the quadrature points in each finite element. A detailed description of the implemented FDM will be presented in the next section.

Overall, the Neighbored Element Method (NEM) combines the FEM and the FDM solution, in a staggered manner, i.e., for each load step, the displacement field $\mathbf{u}$ is computed via a converged Newton–Raphson loop with the field variables p and q kept constant. Alternating, the field variables are then solved for each load step via a separate (linear) equation system while keeping the displacements frozen. This approach is in contrast to the solution in [32], in which the equation for p as well as q in its weak form was approximated by finite element shape functions and discretized as additional degrees of freedom within the FEM.

The NEM was already applied in a similar way in [42] based on the works in [43,44]. However, in these works, a transient partial differential equation was solved via an explicit time discretization, which allowed for a local solution of the according field variables. In the presented work, the equation for the field variable p and q in their strong form do not contain any time derivatives and will be solved directly by means of a (linear) equation system. To directly incorporate the homogeneous Neumann boundary in said equation system, the point cloud for the FDM has to be divided in inner points in Ω_{void} and points on the boundary $\partial\Omega_{\text{void}}$ with their according normal surface normal vectors. Thereby, only the void domain has to be identified $\Omega = \Omega_{\text{void}} \cup \Omega_{\text{solid}}$, where $\Omega_{\text{void}}|_{\chi=0}$. As mentioned before, we choose the quadrature points of the FEM mesh as inner points for the FDM to circumvent additional interpolation schemes between the two discretization. As boundary points, we choose the according quadrature point on the surface elements of the FEM mesh. Both, inner and outer points, are treated slightly different in their respective computation of the differential operators which will be explained in the two following Sections 3.2 and 3.3. The resulting equation system is derived in Section 3.4

3.2. Treatment of inner quadrature points

To approximate spatial differential information on a field variable v in a point of interest $\mathbf{x}_0 = (x_0, y_0)^T \in \Omega_{\text{void}} \setminus \partial\Omega_{\text{void}}$, a Taylor series from that reference point $\mathbf{x}_0$ with spatial coordinates to neighboring points $\mathbf{x}_i \in \Omega_{\text{void}}$ is defined by

$$v|_{\mathbf{x}_i} = v|_{\mathbf{x}_0} + \frac{\partial v}{\partial x} \Big|_{\mathbf{x}_0} \Delta x_i + \frac{\partial v}{\partial y} \Big|_{\mathbf{x}_0} \Delta y_i + \frac{1}{2} \left[\frac{\partial^2 v}{\partial x^2} \Big|_{\mathbf{x}_0} \Delta x_i^2 + \frac{\partial^2 v}{\partial y^2} \Big|_{\mathbf{x}_0} \Delta y_i^2 + 2 \frac{\partial^2 v}{\partial x \partial y} \Big|_{\mathbf{x}_0} \Delta x_i \Delta y_i \right] + \mathcal{O}(|h|^3) \quad (17)$$

with the spatial increments

$$\begin{bmatrix} \Delta x_i \\ \Delta y_i \end{bmatrix} = \mathbf{x}_i - \mathbf{x}_0. \quad (18)$$

For the computation of the Laplacian, the second unmixed spatial derivatives at a point of interest have to be computed, i.e., $\frac{\partial^2 v}{\partial x^2} \Big|_{\mathbf{x}_0}$ and $\frac{\partial^2 v}{\partial y^2} \Big|_{\mathbf{x}_0}$. At least five neighboring points are required to solve Eq. (17), as all first and second derivatives are assumed as unknowns. By applying a Taylor-series to a total number of n neighbors, a equation system in the form

$$\underbrace{\begin{bmatrix} \frac{1}{2} \Delta x_1^2 & \frac{1}{2} \Delta y_1^2 & \Delta x_1 \Delta y_1 & \Delta x_1 & \Delta y_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{1}{2} \Delta x_n^2 & \frac{1}{2} \Delta y_n^2 & \Delta x_n \Delta y_n & \Delta x_n & \Delta y_n \end{bmatrix}}_{\tilde{\mathbf{A}}} \cdot \underbrace{\begin{bmatrix} v_{,xx} \\ v_{,yy} \\ v_{,xy} \\ v_{,x} \\ v_{,y} \end{bmatrix}}_{\tilde{\mathbf{m}}} = \underbrace{\begin{bmatrix} v_1 - v_0 \\ \vdots \\ v_n - v_0 \end{bmatrix}}_{\tilde{\mathbf{b}}} \quad (19)$$

can be formulated. Here, the matrix $\tilde{\mathbf{A}}$ summarizes the respective spatial increments of the neighboring points listed row-wise. The column vector $\tilde{\mathbf{m}}$ lists the derivatives to be determined $(\cdot)_{,j} = \frac{\partial(\cdot)}{\partial j}$, while $\tilde{\mathbf{b}}$ contains the differences of the function values between the value in the reference point and its neighbors.

In order to calculate the derivatives using the neighborhood relations, specific conditions regarding the spatial distribution of the neighbors around the reference point must be satisfied which specify the stability and solvability of the equation system (19). For more details, we refer to [42]. To avoid ill-conditioned equation systems, neighbors should be equally distributed around the point of interest and more than the minimum of five neighboring points should be included. The simplest way to fulfill both requirements is to take all neighboring data points in a specified radius around the point of interest and thus defining an overdetermined system of Eq. (19). To this end, [42] introduced a weighted least square (WLS) approach is applied, so that an approximation of the system of equations is achieved:

$$\mathbf{m} = \underbrace{(\tilde{\mathbf{A}}^T \cdot \mathbf{W} \cdot \tilde{\mathbf{A}})^{-1} \cdot \tilde{\mathbf{A}}^T \cdot \mathbf{W} \cdot \tilde{\mathbf{b}}}_{=: \tilde{\mathbf{D}}_{\mathbf{v}}} \quad (20)$$

The matrix $\tilde{\mathbf{D}}_{\mathbf{v}}$ is the pseudo inverse of the matrices $\tilde{\mathbf{A}}$ with the diagonal matrix $\mathbf{W}$ containing the weights of all included neighboring points. The weighting matrix $\mathbf{W}$ is formed from the respective weightings $w(d_i)$ of the neighboring points to the reference

point, whereby these are based on the Euclidean distance $d_i = \|\mathbf{x}_i - \mathbf{x}_0\|_2$.

$$\mathbf{W} = \begin{bmatrix} w(d_1) & 0 & 0 & \dots & 0 \\ 0 & w(d_2) & 0 & \dots & 0 \\ 0 & 0 & w(d_3) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & w(d_n) \end{bmatrix} \quad \text{with} \quad w(d_i) = \exp \left(-\frac{1}{2} \left(\frac{4d_i}{\sqrt{\beta^*}} \right)^2 \right) \quad (21)$$

The weighting function $w(d_i)$ was introduced in [42] and fulfills the necessary properties by which neighboring points with large distances exert a lower weighting on the reference point. The independent parameter β^* controls the decay of the exponential function and thus the distribution of the weights ($0 \leq w(d_i) \leq 1$) on all neighboring points. We set this parameter equal to the square of the search radius for neighbors around the reference point, which will be chosen accordingly to the mesh of the specific discretized boundary value problem.

In the resulting matrix $\check{\mathbf{D}}_{\nabla}$, the first two rows $(\check{\mathbf{D}}_{\nabla})_{1,i}$ and $(\check{\mathbf{D}}_{\nabla})_{2,i}$ yield the information of the parts for the computation of the Laplacian, i.e., the unmixed second spatial derivatives, whereby the respective neighborhood relations are contained in the columns.

3.3. Treatment of boundary quadrature points

So far, only points within the area Ω_{void} were treated. However, the Neumann boundary condition of the gradient in Eqs. (13) and (16) have to be fulfilled on the boundary $\partial\Omega_{\text{void}}$. For this purpose, we also include the quadrature points on the boundaries of the domain as additional data points for the field variables. Thus, boundary points can also be applied as neighbors in the equation system (19). However, since the Neumann boundary condition has to be fulfilled on these boundary points, the Neumann boundary condition will be directly included as additional equation. This is expressed by an additional row in the matrix $\check{\mathbf{A}}$, which is only assigned at the points located on the boundary $\partial\Omega_{\text{void}}$. The outwards pointing surface normal in the (boundary) reference point is denoted by $\mathbf{n}_0 = (n_x, n_y)^T$. In turn, since the computation of the deformation gradient at the boundary points requires to solve Eqs. (13) and (16) which can become inconvenient, we assume the function value in the boundary points of interest v_0 to be unknown. The enhanced equation system for boundary points thus reads

$$\underbrace{\begin{bmatrix} \frac{1}{2}\Delta x_1^2 & \frac{1}{2}\Delta y_1^2 & \Delta x_1^2\Delta y_1^2 & \Delta x_1 & \Delta y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{1}{2}\Delta x_n^2 & \frac{1}{2}\Delta y_n^2 & \Delta x_n^2\Delta y_n^2 & \Delta x_n & \Delta y_n & 1 \\ 0 & 0 & 0 & n_x & n_y & 0 \end{bmatrix}}_{\check{\mathbf{A}}} \cdot \underbrace{\begin{bmatrix} v_{,xx} \\ v_{,yy} \\ v_{,xy} \\ v_{,x} \\ v_{,y} \\ v_0 \end{bmatrix}}_{\check{\mathbf{m}}} = \underbrace{\begin{bmatrix} v_1 \\ \vdots \\ v_n \\ 0 \end{bmatrix}}_{\check{\mathbf{b}}}, \quad (22)$$

which includes v_0 as additional unknown in the vector $\check{\mathbf{m}}$. Building the system of equation in (22), only inner points are considered as neighbors for the boundary point. Since boundary points are predestined to coincide on one line in a spatial direction, this can lead to a singularity of the matrix $\check{\mathbf{A}}$, or its condition number will increase at least [42]. With the matrix $\check{\mathbf{A}}$ and the weighting matrix $\mathbf{W}$ from Eq. (21), the pseudoinverse matrix $\check{\mathbf{D}}_{\nabla}$ can now be calculated analogously to Eq. (20) as

$$\mathbf{m} = \underbrace{(\check{\mathbf{A}}^T \cdot \mathbf{W} \cdot \check{\mathbf{A}})^{-1} \cdot \check{\mathbf{A}}^T \cdot \mathbf{W} \cdot \check{\mathbf{b}}}_{=:\check{\mathbf{D}}_{\nabla}}. \quad (23)$$

In the resulting matrix $\check{\mathbf{D}}_{\nabla}$, the last row $(\check{\mathbf{D}}_{\nabla})_{6,i}$ yields the information for the computation of the function values on the boundary points so that the homogeneous Neumann boundary conditions are fulfilled.

3.4. Equation system for the field functions for regularization

The construction of a system of equations for the function values of inner points denoted by $\cdot$ and the boundary points denoted by $\cdot$ is explained in this section. The discretized Laplacian from Eq. (20) has to be computed for the solution of the differential equations (13) and (16) while the discretized computation of the unknown function values in the boundary point is given directly by Eq. (23). Both set of equations are assembled into one governing matrix $\mathbf{G}$, which has the size $(\check{N} \times \check{N})$ corresponding to the total number of inner quadrature points $\check{N}$ and outer quadrature points $\check{N}$ with $\check{N} = \check{N} + \check{N}$. As mentioned before, the neighbored information for an inner point is given in the rows $(\check{\mathbf{D}}_{\nabla})_{1,i}$ and $(\check{\mathbf{D}}_{\nabla})_{2,i}$. Therein, each entry refers to one neighbor i of the reference point k . Now, this information will be stored in the corresponding row k of the governing matrix, whereby the column i refers the global number of the neighbor point, i.e.,

$$(\mathbf{G})_{k,i} = (\check{\mathbf{D}}_{\nabla})_{1,i} + (\check{\mathbf{D}}_{\nabla})_{2,i}. \quad (24)$$

For the entry of the reference inner point itself, we evaluate the sum of the rows.

$$(\mathbf{G})_{k,k} := - \sum_{i=1}^n \left((\check{\mathbf{D}}_{\nabla})_{1,i} + (\check{\mathbf{D}}_{\nabla})_{2,i} \right). \quad (25)$$

The boundary points are treated in the same manner. Resulting from the neighbored information for a boundary point given in $(\bar{\mathbf{D}}_{\nabla})_{6,i}$, each entry i will be stored in the corresponding column of the governing matrix, whereby the row k refers to the current reference point, i.e.,

$$(\mathbf{G})_{k,i} = (\bar{\mathbf{D}}_{\nabla})_{6,i}. \quad (26)$$

Accordingly, the entry of the reference boundary point itself reads

$$(\mathbf{G})_{k,k} := - \sum_{i=1}^n (\bar{\mathbf{D}}_{\nabla})_{6,i}. \quad (27)$$

By counting the global number of inner quadrature points first and subsequently the number of boundary quadrature points, the governing matrix $\mathbf{G}$ naturally arises as partition in the form

$$\mathbf{G} = \begin{bmatrix} \check{\check{\mathbf{G}}} & \check{\bar{\mathbf{G}}} \\ \check{\bar{\mathbf{G}}} & \check{\check{\mathbf{G}}} \end{bmatrix} \quad (28)$$

Therein, the partition $\check{\check{\mathbf{G}}}$ stores only the neighbored information between inner points. Likewise, the partition $\check{\bar{\mathbf{G}}}$ contains only neighbored information between boundary points. Since boundary points do not have another boundary points as neighbors, this partition only has entries on its main diagonal. Furthermore, the partitions $\check{\bar{\mathbf{G}}}$ and $\check{\check{\mathbf{G}}}$ contain the neighbored information from outer on inner and inner on outer points, respectively.

For the computation of the Laplacian, the function values in boundary points have to be known. In [42], this was done in a “spatially staggered” manner for an explicit time discretization: first, matrix $\check{\check{\mathbf{G}}}$ was multiplied with the function values of the inner points from the previous time (load) step, i.e., explicit time discretization, to compute the function values in the outer points:

$$[\bar{v}_i] = \check{\check{\mathbf{G}}} \cdot [\check{v}_i]. \quad (29)$$

In a second computation step, i.e., “spatially staggered”, the Laplacian for the inner points was computed by multiplying the combined vectors for the function values from the previous time (load) step and the function values in the boundary points computed in the first step with the upper partition of $\mathbf{G}$:

$$[\nabla^2 \check{v}_i] = \begin{bmatrix} \check{\check{\mathbf{G}}} & \check{\bar{\mathbf{G}}} \end{bmatrix} \cdot \begin{bmatrix} [\check{v}_i] \\ [\bar{v}_i] \end{bmatrix}. \quad (30)$$

Since the differential equation at hand do not contain any transient terms, we combine the two operations (29) and (30) via a multiplication into a condensed system matrix:

$$\left(\begin{bmatrix} \check{\check{\mathbf{G}}} & \check{\bar{\mathbf{G}}} \\ \check{\bar{\mathbf{G}}} & \check{\check{\mathbf{G}}} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \check{\bar{\mathbf{G}}} & \mathbf{0} \end{bmatrix} \right) = \begin{bmatrix} \check{\check{\mathbf{G}}} \cdot \mathbf{I} + \check{\bar{\mathbf{G}}} \cdot \check{\bar{\mathbf{G}}} & \mathbf{0} \\ \check{\bar{\mathbf{G}}} & \mathbf{0} \end{bmatrix} =: \begin{bmatrix} \mathbf{L} & \mathbf{0} \\ \check{\bar{\mathbf{G}}} & \mathbf{0} \end{bmatrix}, \quad (31)$$

whereby the reformulated partition reads

$$\mathbf{L} = \check{\check{\mathbf{G}}} \cdot \mathbf{I} + \check{\bar{\mathbf{G}}} \cdot \check{\bar{\mathbf{G}}}. \quad (32)$$

Finally, we derive the operator matrix $\mathbf{L}$, whose size $(\check{N} \times \check{N})$ now corresponds to the total number of inner quadrature points. This matrix enables the computation of the Laplacian in every discretized inner point via

$$\nabla^2 \check{v} = \mathbf{L} \cdot [\check{v}_i], \quad (33)$$

while fulfilling the homogeneous Neumann boundary condition inherently which are given by Eqs. (12) and (15). By defining the mesh of the void domain, its boundary is known and can be incorporated by the described scheme above. Even complex geometries can be considered as long as the boundary of the void domain is defined. For implementing boundary conditions other than the homogeneous Neumann boundary conditions, a reformulation of Eq. (31) is necessary, i.e., for multiphysics approaches [36]. The reformulation has to consider the given gradient terms on the boundary such that their values are incorporated in the neighbored information. However, since the FDM is defined on a spatial discretization of the quadrature points, also a non-periodic boundary condition could be applied. According to the strong form of the field functions for the regularization given in Eqs. (13) and (16), the discretized Laplacian can now be implemented by the system matrix $\mathbf{L}$. Since the system matrix is based on the discretization of the inner quadrature points, we will discretize the field variables p and q analogously. Therefore, we write the field variables as vector lists $\mathbf{p}$ and $\mathbf{q}$.

$$\mathbf{p} = [p_1, p_2, \dots, p_n]^T \quad \mathbf{q} = [q_1, q_2, \dots, q_n]^T \quad (34)$$

These vectors contain the value of each referenced discretized inner point $i \in \{1, \dots, \check{N}\}$. Analogously, the rotation angles and Jacobians in each discretized point can be formulated as vector lists as well.

$$\boldsymbol{\Phi} = [\tan(\varphi(\mathbf{F}_1)), \tan(\varphi(\mathbf{F}_2)), \dots, \tan(\varphi(\mathbf{F}_n))]^T \quad \mathbf{J} = [J_1, J_2, \dots, J_n]^T \quad (35)$$

Eq. (13) can then be formulated in its discretized form as a system of equations as

$$\mathbf{p} - \frac{\alpha_p}{\beta_p} \mathbf{L} \cdot \mathbf{p} = \boldsymbol{\Phi} \quad \Leftrightarrow \quad \mathbf{L}^p \cdot \mathbf{p} = \boldsymbol{\Phi}, \quad (36)$$

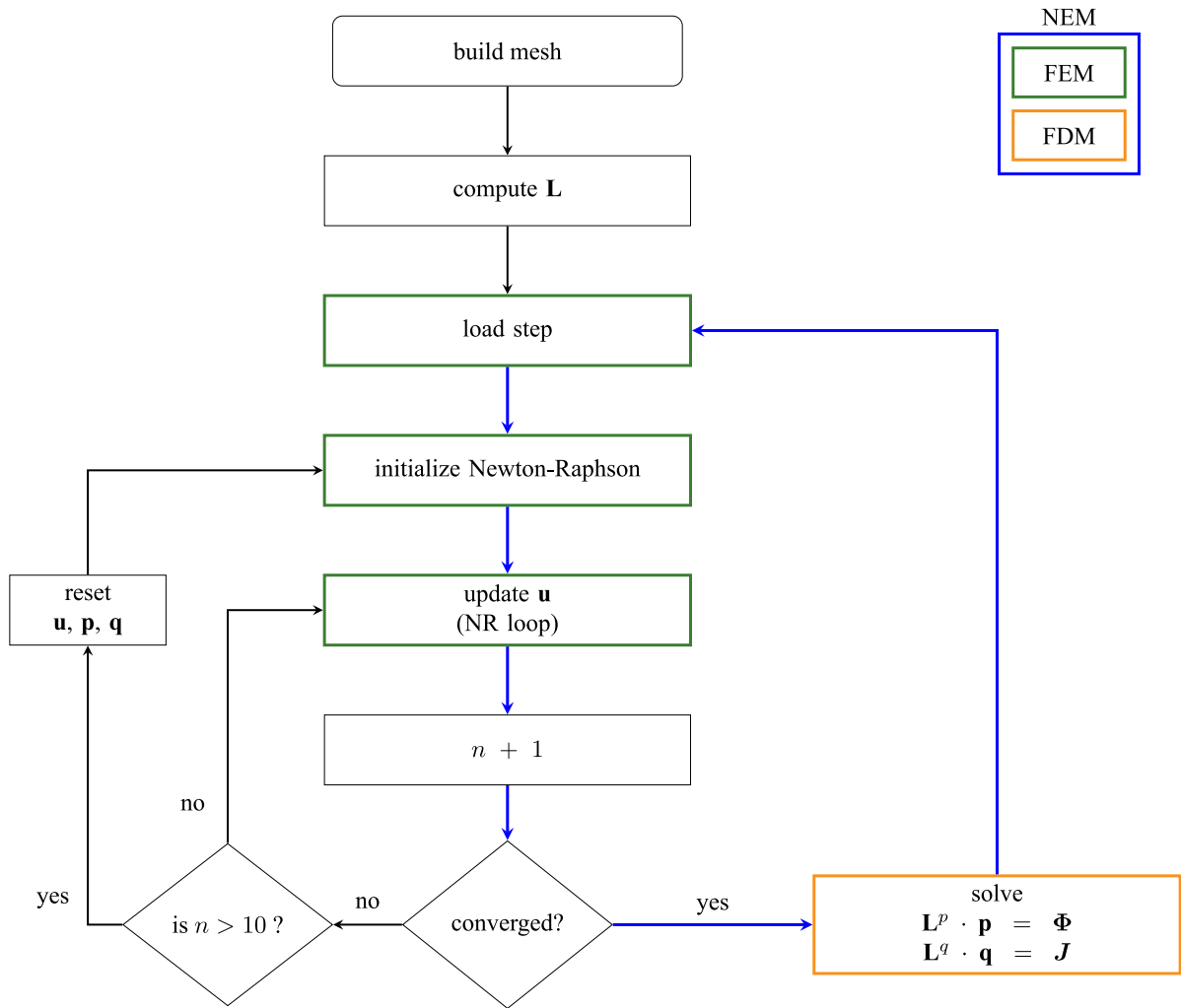


Fig. 1. Flowchart of the solving method, the staggered approach is highlighted.

with the system matrix

$$\mathbf{L}^p := \mathbf{I} - \frac{\alpha_p}{\beta_p} \mathbf{L}. \quad (37)$$

Likewise, Eq. (16) can be formulated equally:

$$\mathbf{q} - \frac{\alpha_q}{\beta_q} \mathbf{L} \cdot \mathbf{q} = \mathbf{J} \quad \Leftrightarrow \quad \mathbf{L}^q \cdot \mathbf{q} = \mathbf{J}, \quad (38)$$

with the system matrix

$$\mathbf{L}^q := \mathbf{I} - \frac{\alpha_q}{\beta_q} \mathbf{L}. \quad (39)$$

As shown in Eqs. (37) and (39), the system matrix $\mathbf{L}^\square$ gains an additional superscript, which indicates the applied field function regularization $\square \in \{p, q\}$. Since both field variables are based on the same discretization, the system matrix $\mathbf{L}^\square$ has to be computed only once, for a given spatial discretization, i.e., a given mesh. Furthermore, as long as the penalty parameters $\alpha_\square$ and $\beta_\square$, with $\square \in \{p, q\}$, remain constant during computations, only the right-hand side of the linear equation system for the update of the field variables p and q changes and according, decompositions can be applied, e.g., LU-decomposition. Fig. 1 shows the solving method of the NEM, which combines the FEM and FDM in a staggered manner.

In this work, the specific implementation of the FDM is performed as followed. Firstly, the geometry of the contact bodies as well as the intermediate domain has to be discretized whereby both together represent the governing mesh. Since only the void domain has to be considered for building the system matrix $\mathbf{L}$, we generate an overlaying mesh which matches the spatial coordinates of

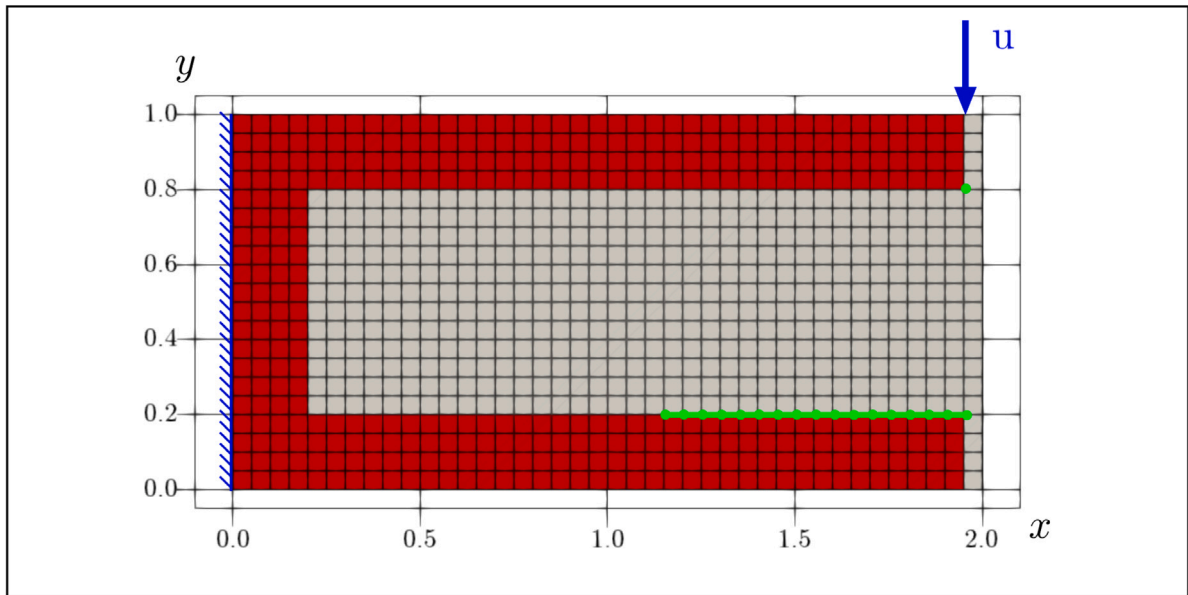


Fig. 2. C-Box in initial configuration and boundary conditions.

the void domain in the governing mesh. It is important to note that this overlaying mesh is only used to build the system matrix $\mathbf{L}$. The boundary of the overlaying mesh can be easily determined and discretized with one spatial dimension lower than the elements within the domain. From here on, the quadrature points within the elements as well as on the boundary are treated as described above, for the inner points and boundary points respectively. Finally, the penalty parameters must be applied to the system matrix, according to Eqs. (37) and (39), resulting in the system matrices $\mathbf{L}^p$ and $\mathbf{L}^q$ for the respective field variables. The penalty parameters $\alpha_{\square}$ and $\beta_{\square}$ with $\square \in \{p, q\}$ have to be tuned manually for the given boundary problem, which follows in the next chapter.

4. Numerical examples

4.1. Common parameters

In this section we discuss the results based on the new approach. We will show that the new approach is able to derive self-contact in structures, but within significantly less computation time compared to the monolithic approach in [32]: herein, each node in the void domain are attributed the field variables p and q as additional degrees of freedom. All simulations are based on two dimensional quadrilateral elements with a first order shape function. By using the first order shape functions, the regularization of the Jacobian could be omitted, according to [32]. Nevertheless, we retain this regularization on every simulation showed here, since by its implementation a smoother distribution of the volumetric deformation of the void elements was observed. The FEM problem will be solved with a classical Newton–Raphson approach, whereby the convergence criteria for the residual norm is given by $1e^{-8}$. The associated parameters for the regularization will be given for each boundary problem separately.

Furthermore, the engineering parameters $\mu = \frac{5}{14}$ and $K = \frac{5}{6}$ maintain constant. The solid elements will be illustrated in red, while gray declare void elements. All simulations were computed using the Julia programming language [45], with the package Ferrite being used for the FEM and the package NearestNeighbors for the search of neighboring points within the NEM.

4.2. Results for the C-Box

The C-Box, originally based on [23], is shown in Fig. 2. The void domain is enclosed from solid beams from three sides while the left outer boarder is fixed and a point-wise displacement load at the upper right corner on the solid domain is applied. An initial displacement $\mathbf{u} = [0.0, -1e^{-3}]$ is applied at the free end of the upper beam. The displacement will be increased by $u_y = -1e^{-3}$ after each converged Newton–Raphson iteration. No body forces are considered. The box has a height of $H = 1.0$ and a length of $L = 2.0$, whereby the thickness of the beams in all sections is $B = 0.2$. The mesh is composed of 800 linear quadrilateral elements. The marked positions on the inner sides of the beams are used to compute the minimal gap during contact. Due to the second-order quadrature rule for the numerical integration, each individual element contains four quadrature points. Two quadrature points per element surface are defined on the boundary region of the void domain $\partial\Omega_{\text{void}}$.

Based on the C-Box problem, we want to compare the presented NEM approach with the monolithic approach from [32]. For each simulation, a total number of 2000 load steps were preset. The chosen parameters for the regularization are given in Table 1.

Table 1
Parameters for the C-Box problem.

Approach	γ	α_p	β_p	α_q	β_q
Monolithic	$1.0e^{-6}$	100.0	10 000.0	100.0	1.0
NEM	$1.0e^{-6}$	1000.0	10 000.0	10.0	0.1

Using the same set of penalty parameters for both approaches did not lead a stable convergence in at least one of the approaches. Due to the different spatial discretization of the parameters p and q and since the parameter α_p and α_q slightly change their reference for the NEM approach, see (37), the actual factor for the Laplacian matrix L resulting from the division of $\alpha_{\square}$ and $\beta_{\square}$ with $\square \in \{p, q\}$ should be accounted for. According to the parameters given in Table 1, the resulting factor for the NEM approach is equivalent to the monolithic approach in the case for the variable q , but significantly smaller in the case for the variable p . However, it will be shown that the results fulfill the entitled contact development.

Some characteristic deformation states of the C-Box problem are shown in Fig. 3, whereby the results of the NEM approach are given on the left hand side, and the results of the monolithic approach are given on the right hand side. It can be seen, that in both cases the contact occurs as expected. At first, the void elements will be contracted homogeneously before the lower tip of the upper beam touches the lower beam. Then, the lower beam starts to bend down due to the contact, while the contact point travels alongside the upper side of the lower beam. It is important to note that no self-penetrations of the solid elements occurs in the contact area during all load steps. With even larger deformations, the contact area extends, by increasing its area to a line to line contact. Its worth to mention, that for the NEM approach the load increment near the end of the deformation state had to be decreased by one tenth to $\frac{u_x}{10}$ to remain stable.

Evidently, both approaches lead to the same general contact behavior and their deformations are approximately identical, except for slight deviations in the deformation of the outer void domain on the right hand side of the box, although they just appearing during the first contact development. However, the gap width gives a more detailed information about the behavior of the void domain. The propagation of the gap width is shown in Fig. 4, whereby the path according to the NEM approach fits the results of the monolithic approach. A closer look after the 500th load step reveals a discrepancy between the results. Herein, the gap width of the NEM approach decreases less over the load steps. This difference only appears when the first contact is made, i.e., the point to line contact. According to the graph in detail in Fig. 5, the maximum difference is in the order of 0.0267, which is considered to be negligible. When it comes to the line to line contact between the bars, closely after the 1250th load step, the path of the NEM approach shows no significant deviation to the monolithic approach. Both paths tends to slightly increase towards the end deformation, close after the 1750th load step. This can be explained with imperfections in the line to line contact, meaning that other points between the bars are in a closer contact. This effect is considered as plausible since points which are closer to each other in a line contact could cause reaction moments which, in turn, can open the gap.

A further comparison is given in Fig. 6, whereby the NEM approach programmed with Ferrite in Fig. 6(a) is compared with an augmented Lagrangian approach computed in ANSYS in Fig. 6(b). Both calculations are set up with the same material model and material parameters. Furthermore, the same amount of load steps was applied in both cases and a classical Newton–Raphson solver was used. Since the augmented Lagrangian approach is based on a predefined contact area, all lines of the inner C-Box are referred to each other to ensure the same contact development, unlike the TMC approach, where the third medium enables the contact. Due to this reason, the third medium, e.g., void elements, are not displayed in the plot in Fig. 6(a) for a better comparison between these approaches. Beside this difference in the formulation of the contact problem, the discretization and boundary conditions given in Fig. 2 are likewise implemented in ANSYS.

The comparison in Fig. 6 shows a similar deformation state resulting from both approaches. In both cases, the upper bar and lower bar undergo a point-to-line contact as well as a line-to-line contact. It is important to note that in both approaches a stable convergence during all load steps was observed. Evidently, the TMC approach is able to provide a similar contact behavior as the augmented Lagrangian approach in ANSYS but without the need for prescribed contact areas.

The investigation of the C-Box, discussed above leads to the following conclusion: in consideration of a slightly different deformation of the void domain as well as the discrepancy in the propagation of the gap width, the introduced NEM approach is able to handle the same contact problem as the monolithic approach. For the cause of the differences, it has to be mentioned that the results depend on the choice of the regularization parameter. Furthermore, due to the staggered manner of the NEM approach, the update of the field variables p and q is carried out after each converged Newton–Raphson step, such that the solution of the field variables is always one step behind the momentary configuration, in contrast to the monolithic approach. According to the stability of the convergence, each load step converged after four Newton–Raphson iterations, except for one of the last load steps in the NEM approach. Hereby, the refinement of the load increment u_x occurred after which the load steps converged after four Newton–Raphsons loops again. Still, the approach exhibit an overall robust convergence.

4.3. Results for the bending bars

We investigate the contact behavior of our method on a second example. As shown in Fig. 7, the setup is defined by two vertical bars which are fixed at their bottoms and one horizontal bar is placed above with an initial gap of 0.1. It is indented that the lower bars are placed asymmetrically to the center of the upper bar, which is referenced by the green dashed line. The purpose of the asymmetric setup is to create two independent contact zones, which undergo different contact behavior. All bars are embedded in

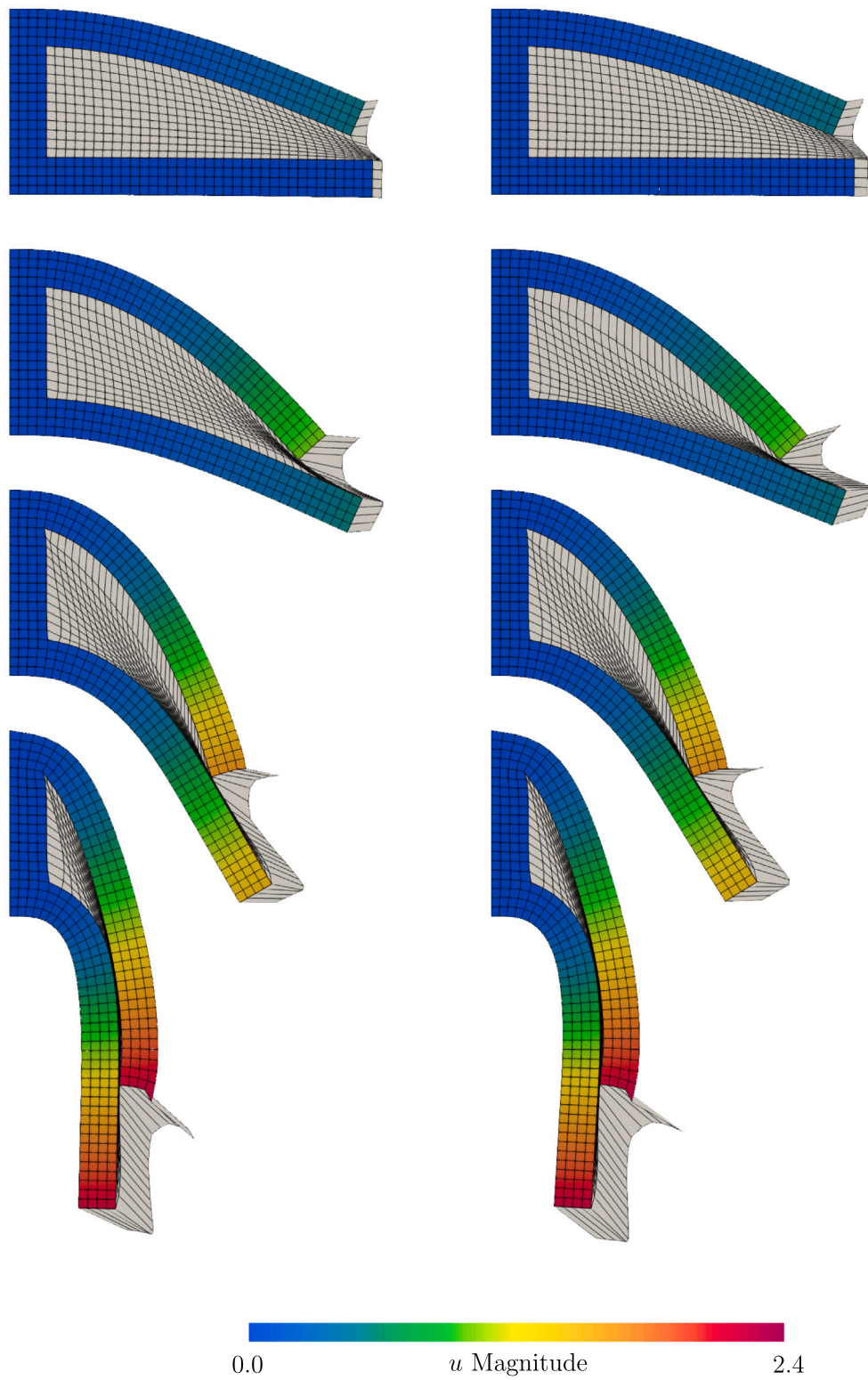


Fig. 3. Comparison of the deformation states of the C-Box, resulting from the NEM approach (left) and monolithic approach (right).

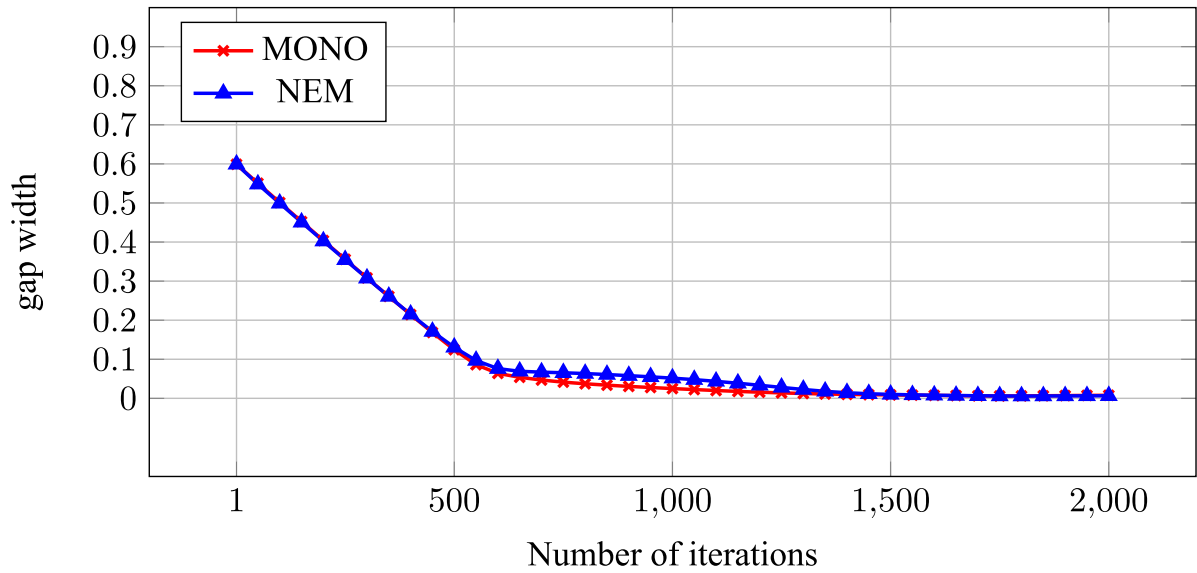


Fig. 4. Comparison of the propagation of the gap width during the simulation.

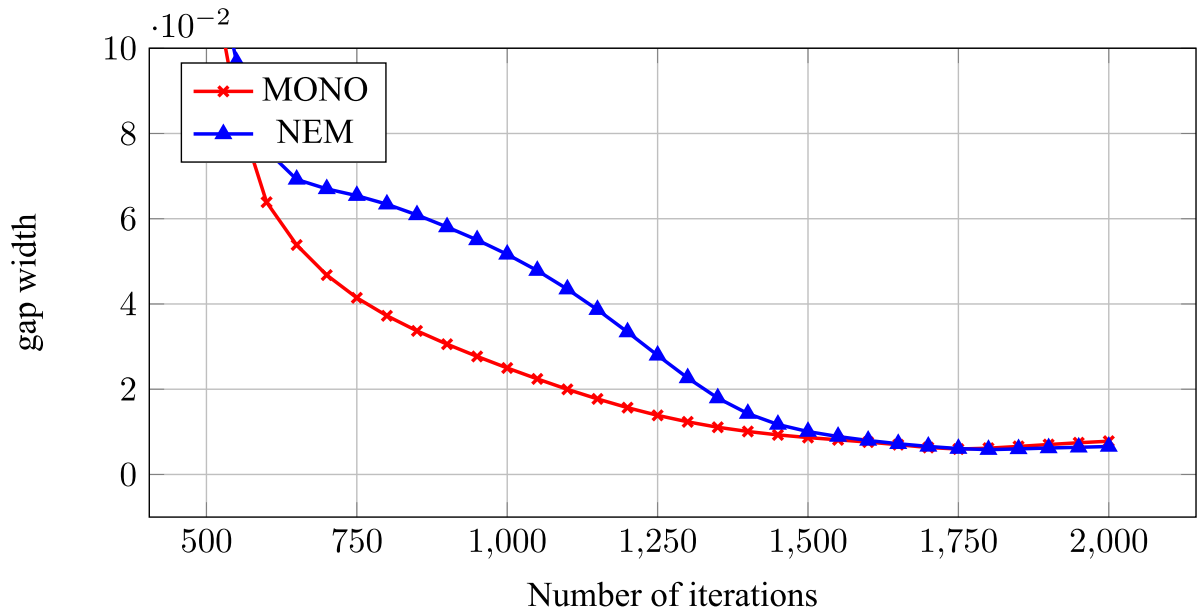


Fig. 5. Comparison of the propagation of the gap width during the end deformation.

void domain, so that a contact could develop at any intersection between the bars. The parameters for the void regularization are listed in Table 2. The entire domain is discretized by a mesh of 3200 quadrilateral elements. As before, each element is described with a second-order geometric quadrature and therewith holds four quadrature points. The initial condition is given by the pointwise applied displacement $\mathbf{u} = [0.0, -1e^{-3}]$ at the center of the upper bar. After each converged load step, the displacement will be increased by $u_y = -1e^{-3}$ whereby no body forces are considered. The incremental loading was used likewise for each approach and a total number of 1000 load steps were preset.

The comparison of characteristic deformation states based on the NEM approach and monolithic approach is given in Fig. 9. Both simulations result in very similar deformations of the bars during all load steps. With the last load step the upper bar undergoes a displacement of $u_y \approx -1.2$, referencing the lower point at the center of the bar, which is more than the half of the size of the domain. However, a closer examination of the last deformation state shows that the results from the NEM approach in Fig. 9(g) tend to a more compressed bending of the upper bar in comparison to the monolithic approach in Fig. 9(h). Since both simulations used the same set of material parameters for the solid, it is to assume that the void domain holds a noticeable influence on the global deformation.

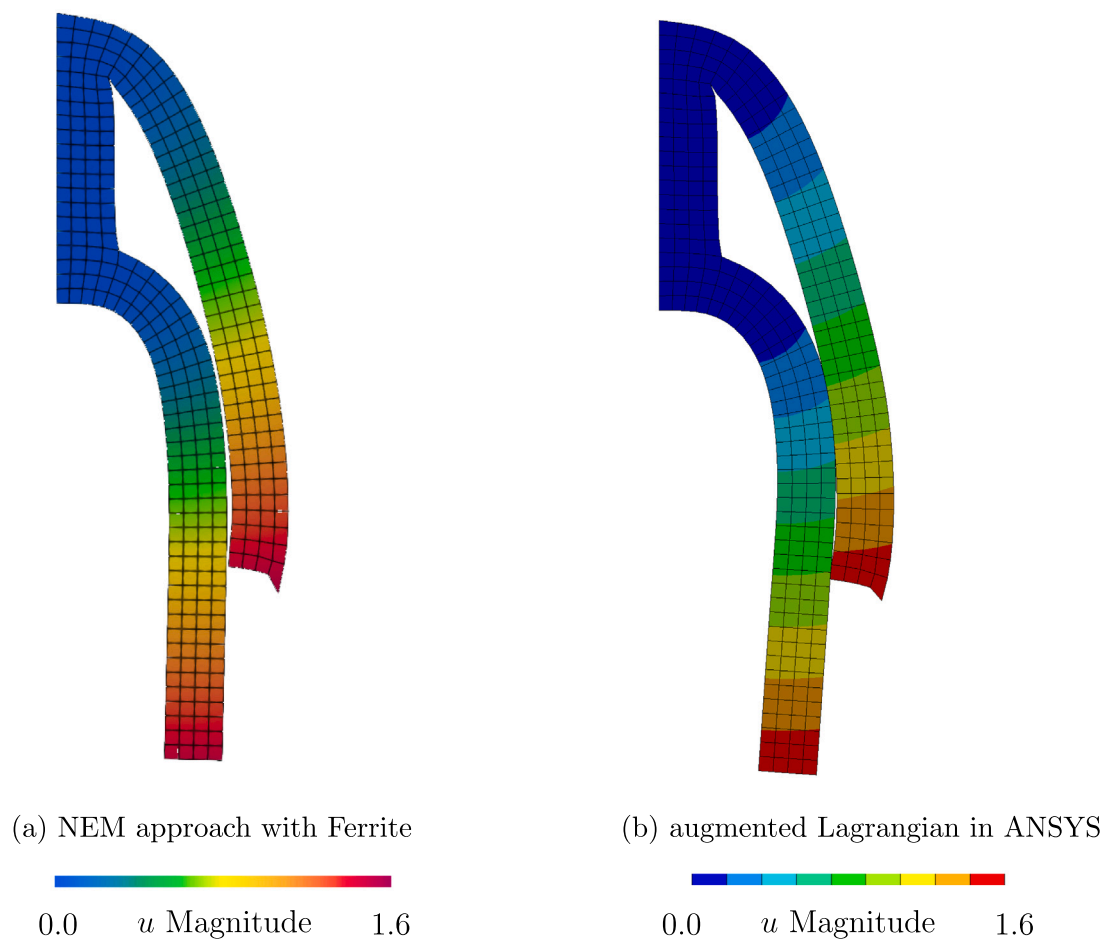


Fig. 6. Comparison of the C-Box problem results between Ferrite (a) and ANSYS (b).

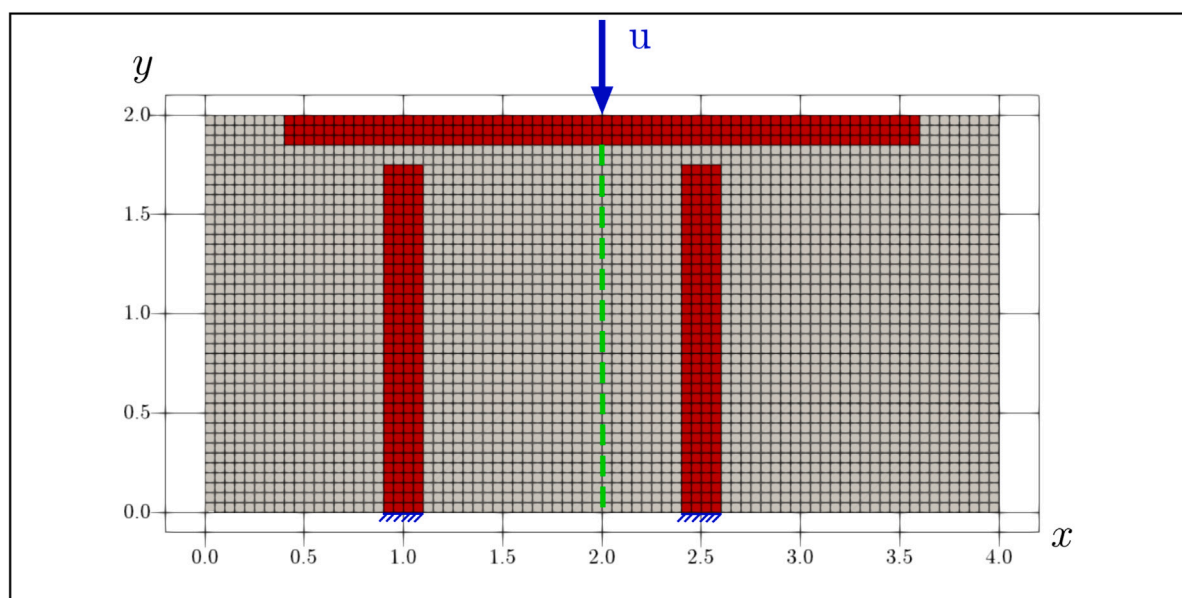


Fig. 7. Initial setup of the bending bar problem.

Table 2
Parameters for the bending bar problem.

Approach	γ	α_p	β_p	α_q	β_q
Monolithic	$1.0e^{-6}$	100.0	10 000.0	100.0	0.1
NEM	$1.0e^{-6}$	1 000 000.0	10 000.0	10.0	0.1

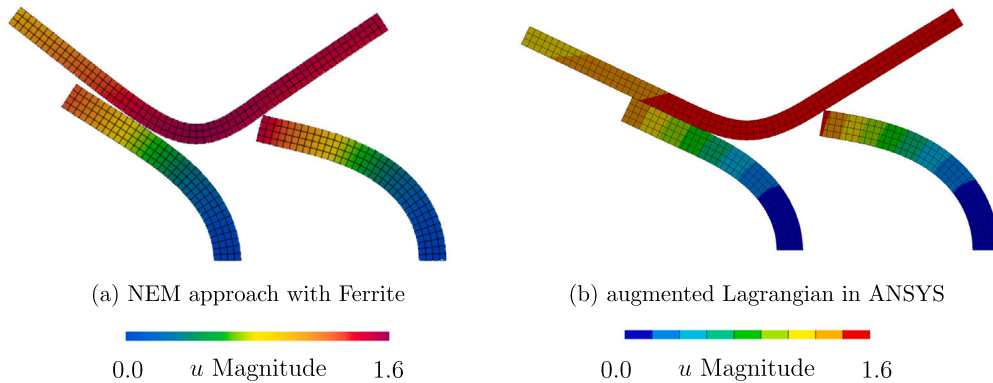


Fig. 8. Comparison of the bending bar problem results between Ferrite (a) and ANSYS (b).

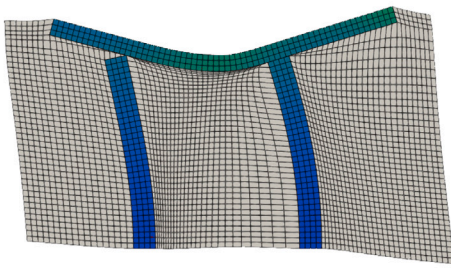
Nevertheless, the void domain shows significant differences between both approaches. Every characteristic deformation state in Fig. 9 presents clearly that the rotation as well as the inflation of the void elements seems to be less regularized in the NEM approach. According to Fig. 9(c) the void elements on the right-hand side undergoing the opposite rotation in comparison to the monolithic approach in Fig. 9(d). Furthermore, an uneven distribution of the element volumes on the right-hand side as well as between the vertical bars is noticeable. These effects become even more significant with further load steps, namely the narrowing of the void elements at the lower right corner and their inflation as direct neighbors of solid elements.

Evidently, these effects are explainable by application of a staggered approach accompanying the different choice of the parameters in Table 2 to grant numerical stability. It has to be emphasized that the actual deformation state of the void elements has no priority in the sense of physical accuracy, i.e., only the deformation and contact in between the solid domains is of interest. The void regions are mere numerical auxiliaries and their explicit behavior is not of any interest. Nevertheless, the NEM approach shows minor differences in the solid deformations compared to the monolithic approach, while stable convergence was observed. It is worth mentioning that within four Newton–Raphson iterations each load step converged. Even for finer meshes, the number of iterations per load steps did not enlarge significantly, only by one to two additional iterations in some load steps.

Another comparison with a conventional method is given in Fig. 8. Herein, the end deformation state of the bending bar problem is plotted for the NEM approach, see Fig. 8(a). The third medium elements, e.g., the void elements are hidden in the plot for a clearer comparison with the results of the augmented Lagrangian approach in Fig. 8(b). Both simulations are based on the same discretization, given in Fig. 7, and material parameters. Furthermore, the step size of the load increment was chosen equally in each simulation. In contrast to the NEM approach, the contact areas in the augmented Lagrangian approach are predefined, whereby every line of each bar was referred to each other, if the development of a contact between them could be possible. Especially, every point, i.e., the upper corners from the vertical bars have to be referred to the line, i.e., the underside if the horizontal bar. This is necessary to ensure that these points do not penetrate the upper bar.

The comparison in Fig. 8 shows a very similar deformation state between the results from the NEM approach and augmented Lagrangian approach. Both approaches are able to develop geometrical point-to-line contact as well as line-to-line contact between the bars. However, the augmented Lagrangian approach shows a wider bending of the upper bar and therewith a more defined line-to-line contact to the left vertical bar. This difference of the deformation state is explainable by the penalty parameters of the NEM approach. It is important to note that the NEM approach shows in fact a sensitivity to the penalty parameters, like the monolithic approach. As a consequence, a tuning of these parameters has to be provided. Still, a very close approximation of the displacement state and the contact development is evidently. Furthermore, the results of the monolithic approach proves this comparison, cf. Figs. 8(b) and 9(h).

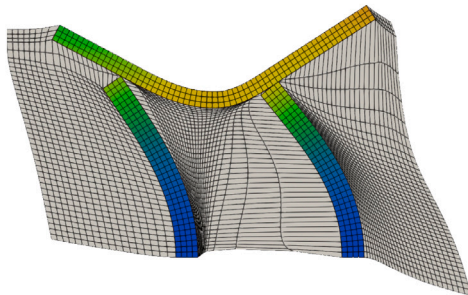
The NEM approach and monolithic approach resulting in the same propagation of the contact. Fig. 9 shows clearly that both methods describe the same contact in each characteristic load step. The right bar gets first into a line contact with the horizontal bar while the left bar undergoes a point contact, i.e., Figs. 9(a) to 9(d). Interestingly, these geometrical contacts change with larger deformations: from a line contact to a point contact and vice versa, compare Figs. 9(c) with 9(e) and Figs. 9(d) with 9(f). The change of contact area was already shown with the C-Box problem, but the bending bars problem prove that the TMC formulation also works for multiple solid bodies which are embedded in one void domain. Nevertheless, with the implementation of the NEM, it will be shown that this approach comes with a reduction of the computation time, while it results in comparable contact behavior as the monolithic approach.



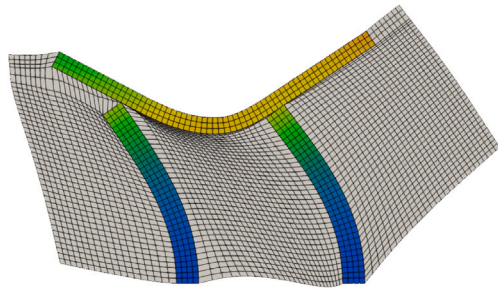
(a) NEM approach at load step 250



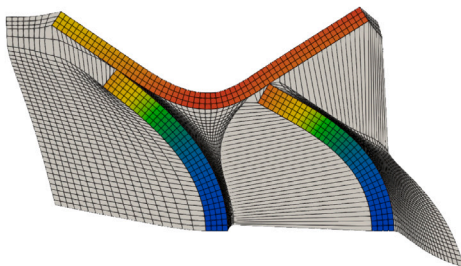
(b) monolithic approach at load step 250



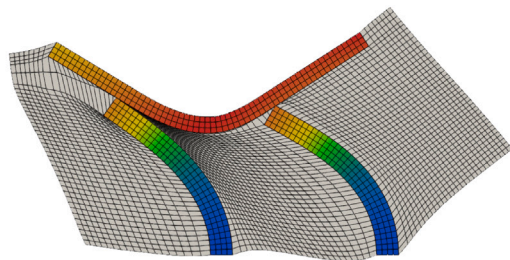
(c) NEM approach at load step 500



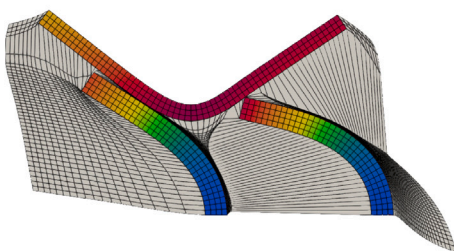
(d) monolithic approach at load step 500



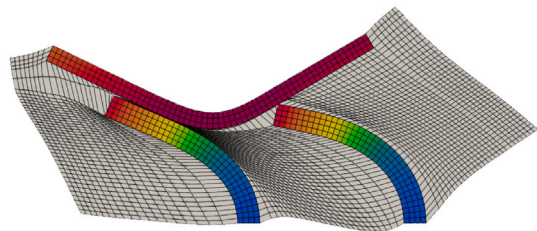
(e) NEM approach at load step 750



(f) monolithic approach at load step 750



(g) NEM approach at load step 1000



(h) monolithic approach at load step 1000

**Fig. 9.** Comparison of the results for the bending bar problem.

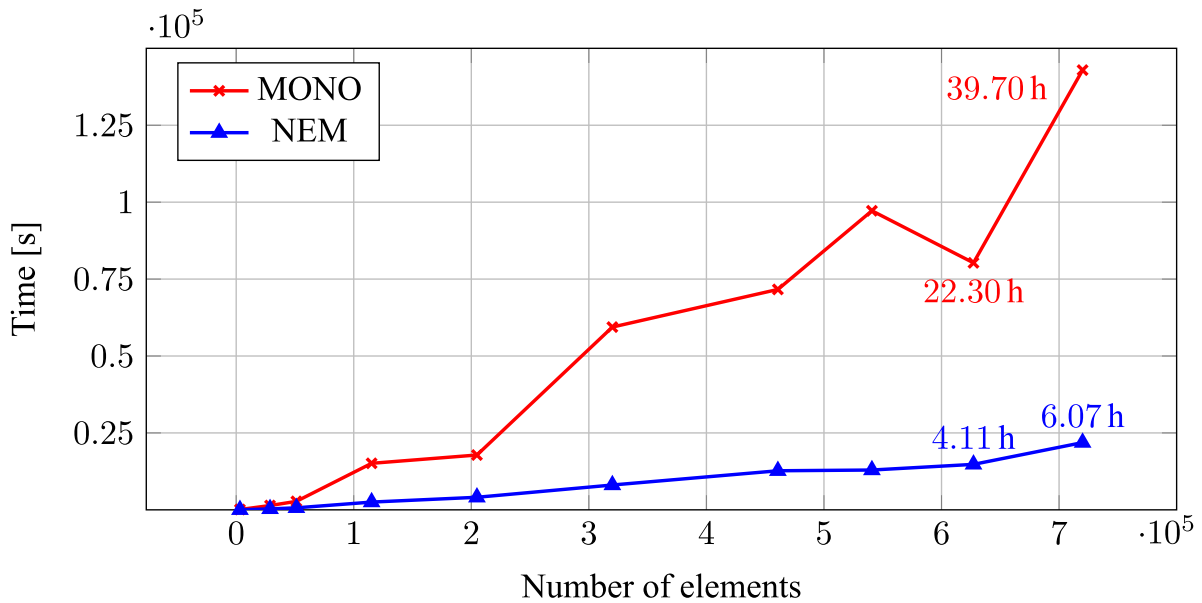


Fig. 10. Comparison of the computation time to solve the displacement $\mathbf{u}$ and update the field variables p and q with the monolithic approach and the NEM approach for ten different mesh refinements.

4.4. Comparison of computing times

The new NEM approach is able to reproduce similar contact developments as the monolithic approach, as discussed before. In this section, it will be shown that the advantage of the NEM approach lays in a reduction of the computation time. Therefore we compare results of the bending bar problem from Section 4.3 with multiple refinements of the mesh. For a better comparison, we refer in Fig. 10 to the computation time which only accounts for the solution of the solver, i.e., the computation of the displacements and the regularization parameters, without accounting for the calculation time of pre-processing or post-processing. Furthermore, we will refer each computation time to the total number of elements, since the total number of degrees of freedom would be misleading due to the varying discretization of the regularization parameters within both approaches: it is important to note that the monolithic approach introduced two additional degrees of freedom in each node of a void element, namely the field variables p and q . Since the NEM approach condense out these field variables as degrees of freedom in each element, they no longer depend on the assembly of the stiffness tangent, but are computed by solving the separated system of equations following from the NEM for each converged load step. Thus, the total number of degrees of freedom cannot be compared directly between the NEM approach and the monolithic approach.

The mesh of the bending bar problem from Fig. 7 was set up for ten different mesh refinements, with the smallest number of quadrilateral elements of 3200 up to the largest number of quadrilateral elements of 720 000. Regarding to the minimal number of nearest neighbors, according to the theory of the NEM, the global search radius for the neighbored area of a quadrature point was preset for each refined mesh, such that the condition of at least five neighbors was fulfilled. As it was mentioned in the origin work of [32], the simulation of the contact problem with the TMC method tends to instabilities when its applied to fine meshes. This effect could also be observed in the present work. However, since we want to evaluate the reduction of the computing time, we adjust a total number of load steps to 150 so that a contact was initiated, but also the simulation of all mesh refinements are stable. Clearly, the computation time will increase by solving more load steps, but by setting the total number of load steps we propose a more reasonable time comparison. As before, the NEM approach and monolithic approach make use of the same load increments. No adaptive refinement of the load increment was observed during the simulation for any given mesh.

An overview of the computation time to solve the displacements $\mathbf{u}$ is given in Fig. 10, whereby the marker indicates the respective approach. Reasonable, the computation time will increase with a larger number of elements. Herein, it is to be assumed that the computation time increases cubically with the number of degrees of freedom for both approaches since both are using the FEM. Nevertheless, the resulting graph of the monolithic approach does not follow an exact exponential path, specially noticeable due to the jump at the last two meshes. By using the standard solver in the program language Julia, it could be the case that the program used another poly-algorithm for the division of the stiffness matrix is at a larger number of degrees of freedom. However, we used the same solver algorithm for both approaches so that the results are comparable. It is remarkable that the NEM approach significantly faster than the monolithic approach for every given mesh. The factor of computational time saving goes up to six and a half, in the presented diagram. Evidently this factor will also increase with finer meshes, i.e., larger numbers elements. This effect would be even more evident for three-dimensional problems. The reason for this saving of computational time results from the staggered method enabled due to the NEM approach. Thereby it is much faster to solve the field variables with an equation systems, defined

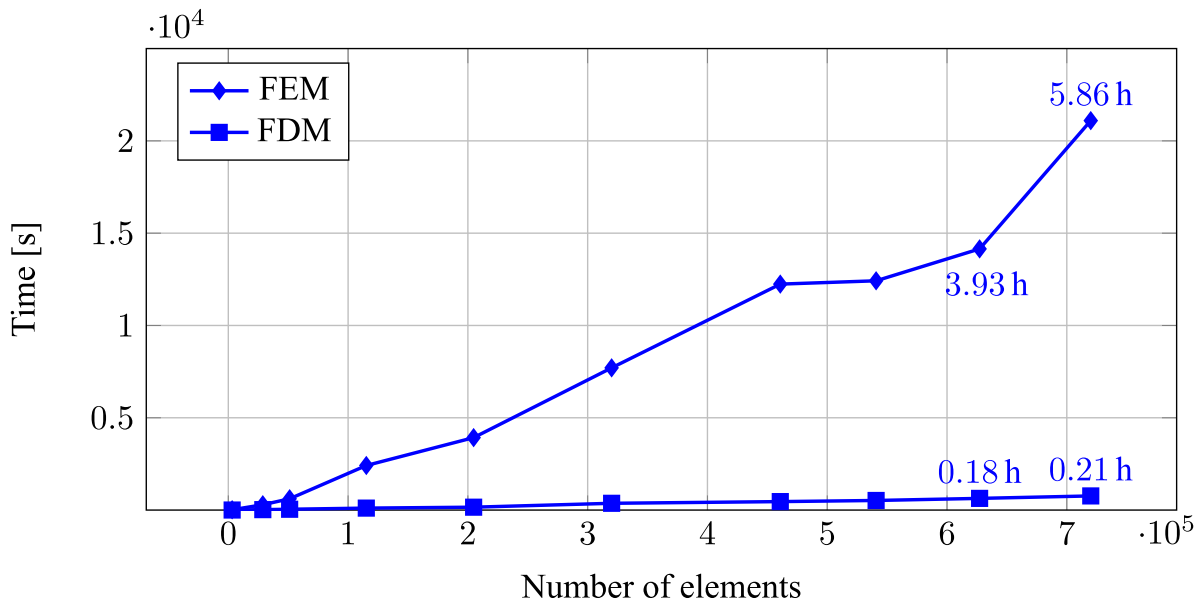


Fig. 11. Comparison of the computation time to solve the displacement $\mathbf{u}$ with the FEM and the field variables p and q with the FDM for ten different mesh refinements.

by the information of neighbored quadrature points than applying each field variable as an additional degree of freedom. This is also shown in Fig. 11 where the computation time to solve the displacement field $\mathbf{u}$ is split into the FEM part and FDM part. The computation of the field variables with the FDM only needs a fraction of time in comparison to the FEM, i.e., up to a factor of 28 for the finest shown mesh. Even for finer meshes, this factor will drastically increase, since the computation time to solve the equation system for the field variables increases linear.

However, staggered approaches tend to be less accurate or unstable for smaller time increments [36]. But in the shown case, even the monolithic approach required smaller load increments to provide stable convergence. The results of both approaches are consistent, except for minor deviations for very large deformation states. Thus, we assume that the drawback of the potential requirement of smaller time increments for the staggered NEM approach is not relevant and that it will be computationally more efficient in most cases.

A further detailed investigation of the computational performance is given in Fig. 12 which presents the total computation time for the Newton–Raphson solver in Fig. 12(a), the total computation time for assembling the residual vector and the stiffness matrix in Fig. 12(b), the total allocated memory per load step in Fig. 12(c), and the allocated memory for the assembling in Fig. 12(d).

Its important to note that a significant saving of computational costs by the NEM approach is evident in all diagrams in Fig. 12. The saving of the total computation time for the Newton–Raphson solver is up to a factor of four, for the finest mesh in Fig. 12(a). In case of the assembling of the residual vector and the stiffness matrix in Fig. 12(b), the computation time reduced by half. It is important to mention that this aspect of the computation part depends on the problem size, or more precisely the number of third medium elements, i.e., the number of void elements. Since the monolithic approach solves the field variables as additional degrees of freedom in the nodes, the residual vector and the stiffness matrix increase in size and so does the necessary computation time. Due to the NEM approach, the field variables are not additionally assembled and the size of the residual vector and the stiffness matrix is determined by the degrees of freedom of the displacement $\mathbf{u}$ only. Consequently, if the governing mesh includes more void domains than solid bodies, the excluded degrees of freedom from the assembling increases. Hence the computation time drops. Furthermore, Figs. 12(c) and 12(d) both show a reduction of the allocated memory by half. However, the comparison of the allocated memory is given here for the sake of completeness. Obviously, this aspect goes along with the problem size and the code performance, whereby the presented approaches are coded in the similar manner. Nevertheless, in [36], a reduction factor of 77 for the allocated memory was achieved in ANSYS, which shows the potential of the NEM approach.

If now the discretization of the domain would be constant, i.e., the number of elements does not change, it is argumentatively reasonable that the time saving depends on the ratio of void to solid elements, knowing that for each void element more field variables have to be assigned. By this means, it can be postulate that the NEM approach will be even faster if the void to solid ratio will increase. This comes with the advantage to apply a larger void domain, i.e., to enable the development of further contact zones without the same increasing computation time as with the monolithic approach.

As mentioned before, the computation times for the pre-process and the post-process are omitted, in all presented comparisons. Obviously the handling of the data for the post-process is subject to the point of interest of the user and thus its associated computation time. However, in the case for the pre-process, it has to be noted that the building of the matrices for the NEM system $\mathbf{L}^\square$ with $\square \forall \{p, q\}$, claims a non-negligible amount of computation time. The necessary computation time depends on the number

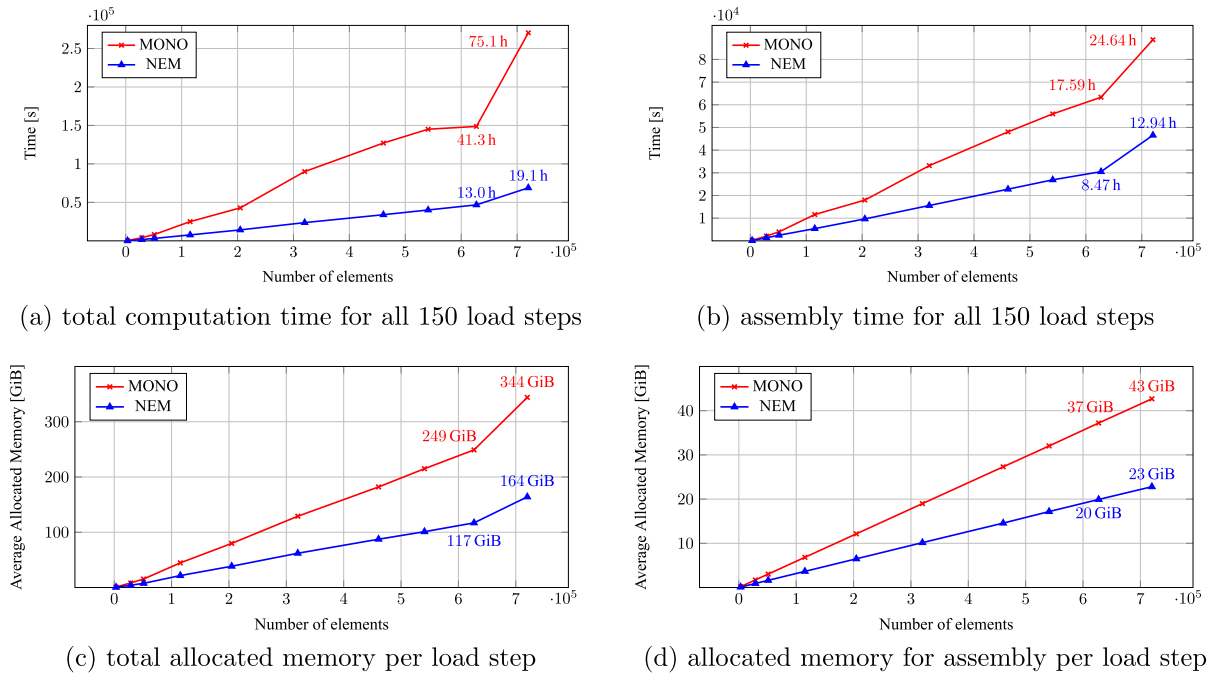


Fig. 12. Comparison of the computational costs component wise for the monolithic approach and NEM approach.

of quadrature points. Thus, the required time is similar to the order of magnitude of the time to assemble the whole stiffness matrix within the FEM approach. However, the matrix $\mathbf{L}^\square$ has to build and decomposed just once for a given spatial discretion and set of regularization parameters, i.e., $\alpha_\square$ and $\beta_\square$ with $\square \in \{p, q\}$. Eventually, it should be emphasizes that the necessary computation time to build and decompose the system matrix $\mathbf{L}^\square$ does not have a major share in the time compared with the whole simulation process.

The time comparison proves the reduction of the computation time by applying the NEM approach. It is important to note that the mesh refinements were equal in both material domains, i.e., the solid and void phase. As explained before, the time reduction is based on the condensation of the field variables from an additional degree of freedom to a separate equation system which can be solved efficiently by a LU-decomposition.

4.5. Results for the ring

The third example we investigate with our method is shown in Fig. 13. Therein, a half of an elastic ring with an outer radius $r = 1.0$ and a thickness of 0.1 is moved towards an elastic fundament. Both solid bodies are colored in red, while the third medium is shown in gray. Marked by the green point and line, the lowest point of the ring and the upper line of the fundament are utilized to determine the gap between them. It important to note that they are only used for the post-process. The initial gap between the bodies is 0.1. Furthermore, both solid bodies share the same material behavior, i.e., the formulation of the strain energy in Eq. (2). The initial condition is given by a point-wise applied displacement $\mathbf{u} = [0.0, -1e^{-3}]$ at the upper part of the ring. After each converged load step, the displacement will be increases by $u_y = -1e^{-3}$ whereby no body forces are consider. Since the upper part of the ring is only able to be displaced in the y direction, the horizontal diameter of the ring is fixed. Consequently, if the ring gets into contact with the fundament it will undergo large deformations, since the structural compliance of the ring is significantly lower against the compliance of the fundament. Furthermore, it is assumed that the ring will snap back if its compressed further onto the fundament. A variation of the penalty parameters is intended to investigate their influence on the stability of the NEM approach and its results.

An detailed investigation of the penalty parameters was already provided by [32]. In case of the NEM approach, the penalty parameters $\alpha_\square$ and $\beta_\square$ with $\square \in \{p, q\}$ have the same effect on the regularization as for the monolithic approach, namely that the field variable p regularizes the rotation of the third medium elements and q regularizes their volumetric change. However, it was observed that the value of the penalty parameters has to be adjusted in order to achieve a similar regularization, i.e., contact development as in the monolithic approach. An excerpt of this is already given in Table 1 for the C-Box and in Table 2 for the bending bar problem. Based on the ring problem, we want to observe the influence of the penalty parameters on the gap width and the deformation state to provide a guideline for a parameter selection. The investigated penalty parameter sets are given in Table 3.

The parameter sets in Table 3 are sampled whereby a significant change of the ratio between both penalty parameter pairs is taken into account. The resulting deformation state is given in Fig. 14, referred to each set of parameters in Table 3. Since all results show a similar minimal gap width, the main aspect of evaluation is the deflection of the ring.

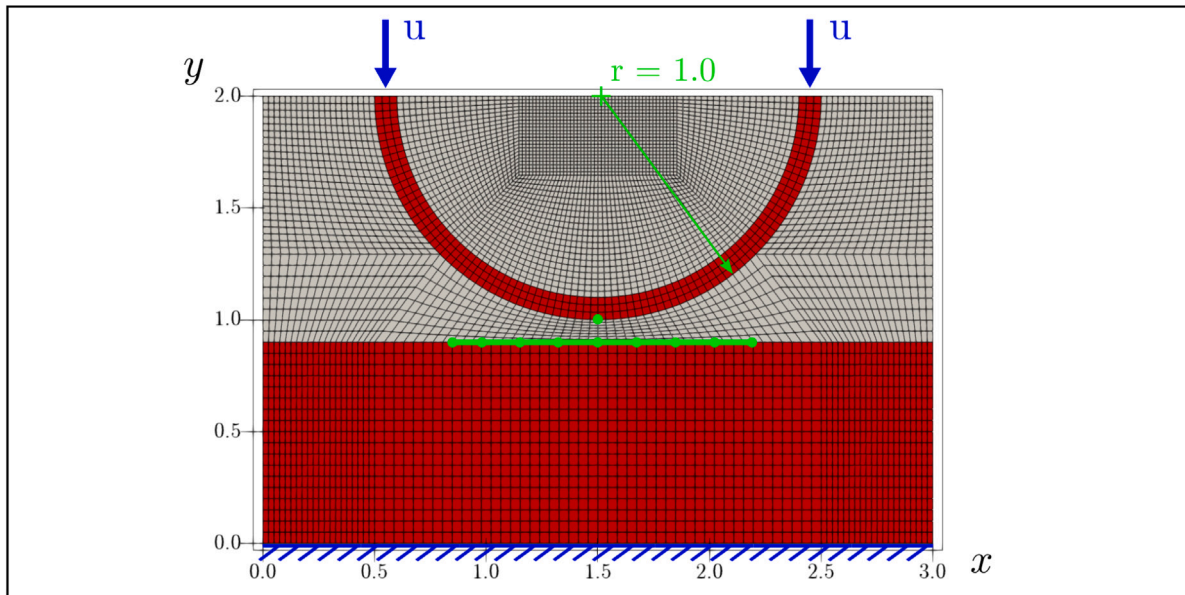


Fig. 13. Initial setup of the ring problem.

Table 3

Parameter sets for the ring problem.

Reference	γ	α_p	β_p	α_q	β_q
Fig. 14(a)	$1.0e^{-6}$	10.0	100.0	10.0	0.1
Fig. 14(b)	$1.0e^{-6}$	1000.0	10 000.0	10.0	0.1
Fig. 14(c)	$1.0e^{-6}$	1 000 000.0	100 000.0	10.0	0.1
Fig. 14(d)	$1.0e^{-6}$	1000.0	10 000.0	$1.0e^{-6}$	10.0
Fig. 14(e)	$1.0e^{-6}$	1 000 000.0	10 000.0	$1.0e^{-5}$	100.0
Fig. 14(f)	$1.0e^{-6}$	1000.0	10 000.0	$1.0e^{-5}$	100.0
Fig. 14(g)	$1.0e^{-6}$	10.0	10 000.0	$1.0e^{-5}$	100.0
Fig. 14(h)	$1.0e^{-6}$	10.0	100.0	$1.0e^{-5}$	100.0

Comparing the deformation states in the Figs. 14(a), 14(b), 14(c), 14(d) and 14(h) shows that the according parameter sets lead to very similar results. All of these five parameter sets enable a proper contact and a lifting of the ring, although the magnitude of deflection is in a smaller order. Comparing the referred parameter sets in Table 3, it is important to note that the ratio of α_q and β_q has not a significant influence on the result, while the values of α_p and β_p are relatively small, cf. Figs. 14(a) and 14(h). Likewise, if the ratio of α_q and β_q is relatively large, while the values of α_p and β_p are of a much higher order, the results seem not to be effected, cf. Figs. 14(c) and 14(d). Nevertheless, applying an even larger ratio of $\frac{\alpha_p}{\beta_p} < 1$ does not lead to better results, as it shows in Fig. 14(e). Thereby, the effect of a snap back was not enabled, although a proper contact was derived. It is clearly to see that the heavy distortions of the void elements cause numerical instabilities. However, the NEM approach is still very sensitive to the chosen penalty parameters, as can be seen by comparing Figs. 14(d) and 14(f). Thereby, the deflection of the ring in Fig. 14(f) is much larger than in Fig. 14(d), while each of their parameter ratios are equal, cf. Table 3. As a result, not only the ratio of the parameters is effecting the result, but especially the value of the penalty parameter β_q , i.e., the regularization of the Jacobian. This is also evident by comparing Figs. 14(f) and 14(g), wherein both results show a large deflection but a smoother distortion of the void elements in Fig. 14(g). Furthermore, the set up in Fig. 14(f) failed after 646 load steps, whereas the setup in Fig. 14(g) converged over all preset 650 load steps. In conclusion, a small ratio of the penalty parameters leads to a larger deflection in the presented example, while the regularization of the Jacobian, i.e., the ratio of α_q and β_q , has a more significant influence on the end deformation state and therefore on the robustness of the NEM approach.

Concluded, it can be reported that the parameter $\gamma = 1.0e^{-6}$ is quite optimal for the presented examples. It was observed that smaller values of γ tend to a penetration of the solid bodies in the contact area and higher values results in a stiffening of the third medium, i.e., the void elements stiffen even before a proper contact is developed. Furthermore, the convergence of the NEM approach was much more sensitive to small changes of the penalty parameters α_q and β_q , i.e., the regularization of the Jacobian. In contrast, a wider variation of the penalty parameters α_p and β_p , according for the regularized rotation, shows less influence on the convergence behavior. Its important to mention that in general a small ratio of each penalty parameter pair, i.e., $\alpha_{\square} < \beta_{\square}$ with $\square \in \{p, q\}$, was leading to a more robust convergence towards the end deformation state. Furthermore, it was found that significant smaller values of α_q have a positive effect on the convergence robustness. Nevertheless, it should be noted that the influence of the selected penalty parameters may vary depending on the type of problem.

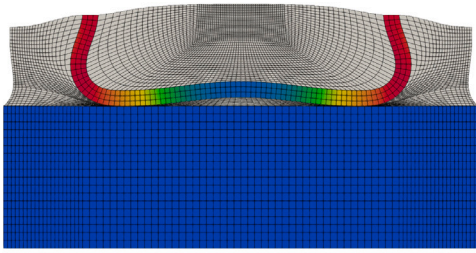
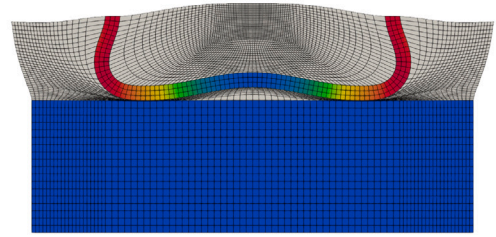
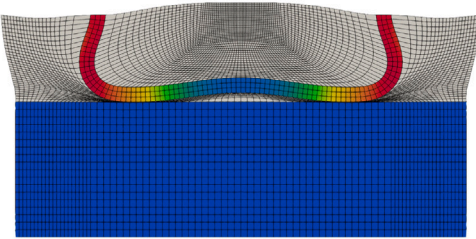
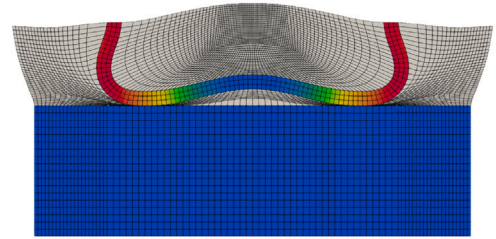
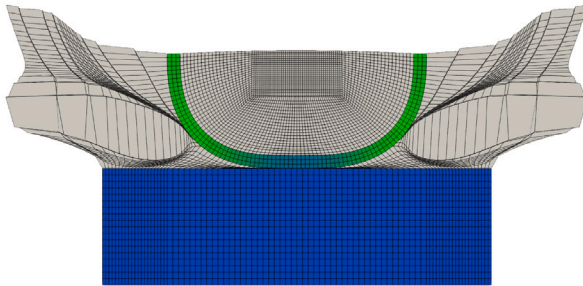
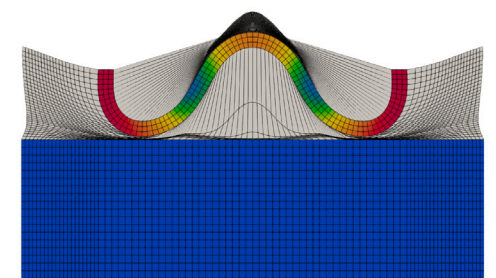
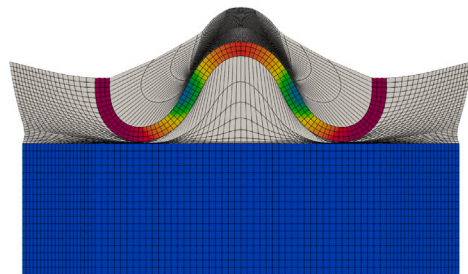
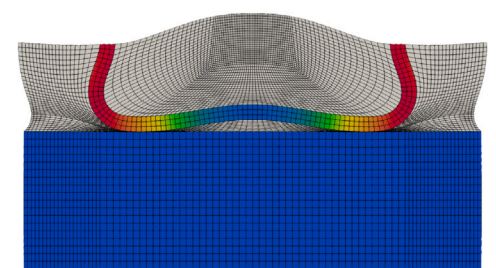
(a) minimal gap = $6.46e^{-4}$; load step 526(b) minimal gap = $6.80e^{-4}$; load step 529(c) minimal gap = $6.87e^{-4}$; load step 523(d) minimal gap = $6.75e^{-4}$; load step 530(e) minimal gap = $6.85e^{-4}$; load step 214(f) minimal gap = $6.74e^{-4}$; load step 646(g) minimal gap = $6.38e^{-4}$; load step 650(h) minimal gap = $6.36e^{-4}$; load step 529

Fig. 14. Comparison of the influence of the penalty parameters on the ring problem, showing each end deformation state according to the given penalty parameters in [Table 3](#).

5. Conclusion

In this work, we introduced a novel numerical approach to the third medium contact (TMC) which introduces additional field variables for regularization [32]. The primary advance of this work is the reduction of computation time for all cases examined by a factor of up to four depending on the spatial discretization while contact behavior remains comparable to the previous approach. To this end, the Neighbored Element Method (NEM) was applied to determine the unknown field variables with a spacial discretization, i.e., the displacement field remains to be discretized by means of a finite element discretization whereas the field functions for regularization are discretized within the quadrature points of the FEM mesh. The displacement field and the field functions are solved in a staggered scheme in which the field functions are solved via a generalized finite difference scheme. The system matrix for the solution of the field functions only depend on the spatial discretization and constant numerical parameters regarding the regularization and can be thus build and decomposed once for a given problem yielding an overall very efficient numerical solution. For this purpose, we had to introduced direct solution within the FDM, since previous applications only assumed explicit time discretization for transient partial differential equations.

Furthermore, it was shown that the NEM approach leads to the same results as the monolithic approach while it was likewise stable under the same conditions. Furthermore, a comparison with an augmented Lagrangian approach carried out in ANSYS shows a very close approximation of the deformation states. As for the monolithic approach in [32], results seem to be rather sensitive to the chosen values for the penalty parameter, accounting for the regularization of the deformation of the void domain, which had to be redefined for the NEM approach. Still, with the given selection of parameters for each problem, we were able to reproduce similar contact behavior as with the monolithic approach and a negligible differences in the resulting gap width. A sampled set of penalty parameters provides a guideline of the influence on the results. However, future work will tackle automated computation of optimal penatly paramters and/or adaptive schemes for more user-friendly application.

The here presented implementation of the NEM approach in the TMC enables a fast computation strategy to simulate frictionless contact. Nevertheless, the selection of parameters has a significant influence on the deformation of the void domain, and with that – although the actual deformation of void is irrelevant – they also have some influence on the contact development. Further research is required on the behavior and optimal determination of the regularization parameters.

One obvious further investigation is the enhancement to 3D elements. In [34], the regularization for 3D elements was achieved by extending the energy term of the regularization by a skew part of the deformation gradient. Thereby, the symmetric skew-like part leads to three components for the regularization which require three additional field variables, instead of just one field variable in case of 2D elements. Herein, the NEM approach represents a promising alternative solution, since additional field variables are not treated as additional degrees of freedom and consequently reduces the computational costs. The extension of a third spatial coordinate on the search of neighbors in the FDM is simple. However, the enhanced regularization terms must be implemented in the extended Hamilton principle subsequently. Since the NEM approach includes the classical FEM approach, other material models can be applied for the solid elements. Furthermore, the adaptation of multiphysics is easy to handle with the NEM, shown in [36].

Last but not least, introduction of friction in the contact area is a comprehensible aspect. Since the contact area is not predefined, a spatial criteria for indicating friction is unsuitable. Rather, an additional strain energy, or respectively a regularization of the shear stiffness of highly compressed void elements, seems to be more suitable for the TMC approach. Inspired by crystal plasticity the work in [26] enabled friction by adding an anisotropic strain energy term. A similar extension of the formulation of internal energy for the presented approach is also conceivable, but as a consequence, the additional energy term must be implemented in the extended Hamilton principle.

CRedit authorship contribution statement

Max von Zabiensky: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Conceptualization. **Dustin R. Jantos:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Philipp Junker:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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